

Optical Integrated Sensing and Communication for 6G Through a Systematic Review of Architectures, Metrics, and Tradeoffs Across Optical Platforms

Fatih Dönmez, Ahmet Altuncu, and Mustafa Namdar

Abstract—Light carries data and signatures of the path and objects it encounters, allowing the same optical infrastructure to support communication and sensing. Optical integrated sensing and communication (O-ISAC) builds on this dual capability across fiber, free space optical, visible light, and light fidelity (LiFi) platforms. Photonic hardware extends the concept to millimeter wave and terahertz systems, while hybrid designs connect physical domains. This breadth motivates comparison of the design insights that extend across platforms and those tied to a particular physical setting. A systematic literature search covering 1 January 2020 through 22 June 2026 identified 227 eligible reports representing 206 unique studies. We organized this evidence by physical platform and coupling mechanism using structured narrative synthesis and a technical assessment developed for this review. The comparison framework connects each result to its task, measurement plane, operating conditions, and validation setting. Across the included corpus, bandwidth, optical power, hardware constraints, and signal design jointly shape communication reliability and sensing performance. These relationships depend on reported conditions and extend beyond a single rate and resolution frontier. Laboratory studies span diverse architectures. Broader field validation, shared benchmarks, and reusable data and code emerge as priorities for stronger comparison and reconstruction. The resulting taxonomy and framework support the design and evaluation of O-ISAC systems for sixth generation (6G) networks.

Index Terms—6G, integrated sensing and communication, metric comparability, optical communication, optical sensing.

I. INTRODUCTION

A. Background and Motivation

Optical communication infrastructure can support data transmission alongside physical measurements, such as vibration along a fiber or receiver position in a room [1], [2]. Optical integrated sensing and communication (O-ISAC) uses this

capability to combine the two functions through shared infrastructure, hardware, or signals [3], [4]. This extends integrated sensing and communication (ISAC) to optical and photonic systems [5]. ISAC is included among the usage scenarios in the International Mobile Telecommunications framework for 2030 and beyond [6]. The opportunity to reuse resources brings a joint design problem, since their operation must account for both communication and sensing requirements.

How these resources are shared depends on the signal path and the sensing task. In fiber and optical wireless systems, light interacts with the sensing medium or reaches the receiver or target [3], [7], [8]. In photonics-enabled wireless systems, optical components can generate the signal, while propagation to the target occurs in the millimeter-wave or terahertz band [9], [10]. Figure 1 illustrates these physical settings and the resources used by both functions. Its common design question is how choices of power and waveform, together with timing and receiver design, affect communication and sensing performance.

Shared operation can introduce a tradeoff or allow one function to assist the other. In one fiber experiment, raising the sensing-probe power while holding communication launch power fixed improved the sensing signal-to-noise ratio but increased the communication bit error rate [1]. In a visible-light prototype, position estimates guided beamforming toward the receiver and improved the communication data rate at the tested locations [2]. The fiber result illustrates a power-dependent tradeoff, whereas the visible-light result shows how sensing information can guide transmission.

A researcher selecting an O-ISAC approach needs to know which integration mechanism fits the application and how it affects both functions. Reported rates and sensing errors become useful for that choice when they are linked to the shared design and its operating conditions. These links can explain why a method is effective in one setting and what must be tested before adapting it to another. A synthesis of the literature should therefore connect physical architectures to joint performance and to the experiments needed for further development.

B. Related Surveys

Existing reviews provide several perspectives on these design questions, through platform-specific analyses and broader views of O-ISAC. A tutorial on optical architectures and waveform allocation uses a common simulation setting to show

This review received no specific financial or nonfinancial support. No funder or sponsor had any role in the design, conduct, analysis, interpretation, manuscript preparation, or decision to submit.

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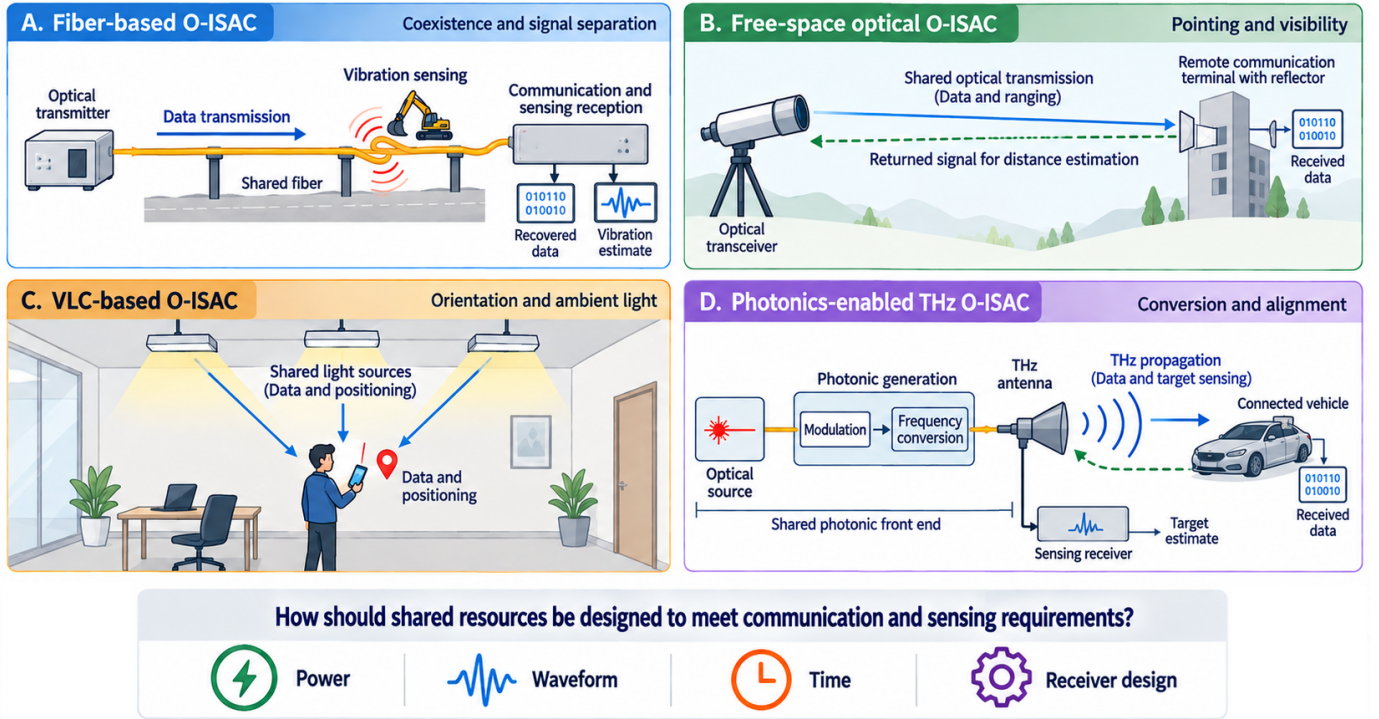


Fig. 1. Representative O-ISAC configurations in fiber, free-space optical, visible-light, and photonics-enabled terahertz systems. Each panel identifies a shared resource, the two functions it supports, and a representative physical constraint. The lower panel highlights power and waveform choices alongside timing and receiver design as common design considerations.

communication error rate and ranging error [11]. A review of lighting, sensing, and communication integration provides performance tables for visible-light and hybrid systems [12]. A waveform-focused review examines photonic millimeter-wave and terahertz demonstrations and compares integrated waveforms through models and experiments [13]. Other focused reviews address optical transmission and estimation, or communication and sensing in fiber networks [14], [15], [16], [17].

Broader surveys connect these technical developments to larger research questions. An evolution-oriented survey traces ISAC across spectrum, network scale, and sensing modalities, including optical and RF–optical systems [5]. A broad O-ISAC review brings together optical platforms, experimental results, waveform design, and applications [4]. These two surveys also discuss performance measures and the need for further evaluation. Together, these works explain design choices, report numerical demonstrations, and identify research directions.

Table I distinguishes the physical coverage and review approach of representative works. Its platform columns identify where the sensing interaction takes place. The last two columns address how studies are selected and how results from different studies are judged comparable. This distinction matters because a controlled waveform comparison and a comparison between independently reported experiments answer different questions. The latter also requires agreement on what was measured and under which conditions.

A further need is to explain when a reported result can guide another design. The difficulty of using common performance

measures across RF and optical systems is discussed in [5], while shared benchmarks are identified as a research need in [4]. These observations motivate comparison rules that connect shared design choices to both performance outcomes and specify the measurement conditions under which results can be compared.

C. Scope and Contributions

This survey examines O-ISAC in guided optical systems, optical wireless links, and photonics-enabled wireless systems, including hybrid configurations. We analyze how shared infrastructure, signals, and design objectives connect communication and sensing, and how these connections shape joint performance.

The contributions address physical integration, performance interpretation, and experimental evaluation.

- **A physical basis for selecting an O-ISAC design.** We distinguish platforms by their sensing paths and identify how hardware, signals, and allocated resources connect the two functions. This explains which integration mechanisms fit a given physical setting and which constraints must be considered when adapting them (Sections II and IV).
- **A common basis for interpreting performance.** We relate waveform and receiver choices, together with resource allocation, to communication rate and reliability alongside sensing accuracy and resolution. We distinguish relationships demonstrated within a study from com-

TABLE I
PHYSICAL COVERAGE AND REVIEW APPROACH OF REPRESENTATIVE RELATED REVIEWS AND TUTORIAL-STYLE SYNTHESSES

Ref. / year	Main focus	Optical platforms			Review approach	
		Fiber	Optical wireless	Photonic wireless	Study selection	Cross-study comparison rules
[11] (2024)	Optical architectures, waveform design, and resource allocation	—	●	—	—	—
[12] (2024)	Lighting–sensing–communication integration and hybrid links	—	●	—	—	—
[13] (2026)	Integrated photonic waveforms and controlled performance comparisons	—	—	●	—	—
[5] (2026)	ISAC evolution across spectrum, networks, and sensing modalities	●	●	●	—	●
[4] (2026)	Optical platforms, experimental demonstrations, and applications	●	●	●	—	●
This survey	Shared-resource performance relations and evidence-based evaluation	●	●	●	●	●

●: structured treatment; ●: selected discussion or an identified research need; —: not developed in the stated sense. Fiber denotes sensing in the fiber, not fiber transport alone; photonic wireless denotes optically generated or processed signals with wireless RF propagation. Only optical-related coverage is marked for [5]. Study selection denotes an explicit search and eligibility procedure. Cross-study comparison rules address metric definitions, measurement points, conditions, baselines, and supporting evidence; they are distinct from a controlled comparison within one model or experiment.

parisons that require compatible metrics, measurement points, and operating conditions (Sections II and V).

- **Evaluation priorities derived from the evidence.** We derive joint tests from the validation limits observed across optical platforms. These tests pair communication and sensing measurements with separate-function baselines under changes in traffic, propagation conditions, or hardware stability (Sections VI and VIII).

Section II introduces the physical and performance concepts used in the review, and Section III describes the review methods. Sections IV through VII examine platforms, joint design, validation, and applications. Section VIII develops the research priorities, and Section IX concludes the article.

II. TECHNICAL FOUNDATIONS OF O-ISAC

The physical interaction that enables sensing also determines how it can share resources with communication. This section follows the signal from that interaction to the receiver, then explains the optical constraints on sharing and the measures used to evaluate both functions. Numerical examples show how a shared setting connects the resulting performance, providing a basis for the architectures and design relationships in Sections IV and V.

A. Signal Paths and Sensing Observations

A communication receiver recovers information encoded by the transmitter. A sensing receiver estimates a property of the target or propagation medium from its effect on the received signal. The two functions can use the same received samples, different observations of one transmission, or separate signals using shared infrastructure. Figure 2 separates signal generation, physical interaction, and receiver processing. This separation locates both the sensing mechanism and the point at which each performance metric is measured.

The sensing observation depends on the physical task. Fiber vibration can change the phase or polarization of a transmitted or backscattered field. Reflective ranging uses a return whose delay depends on the target distance. Visible-light positioning can instead infer receiver location from the intensities of known illumination patterns. These paths lead to different estimators even when they share optical components [1], [2], [18], [19].

Receiver design determines which signal properties are available for estimation. For an intensity-modulated, direct-detection link operating in a linear intensity-channel regime, the electrical observation can be written as

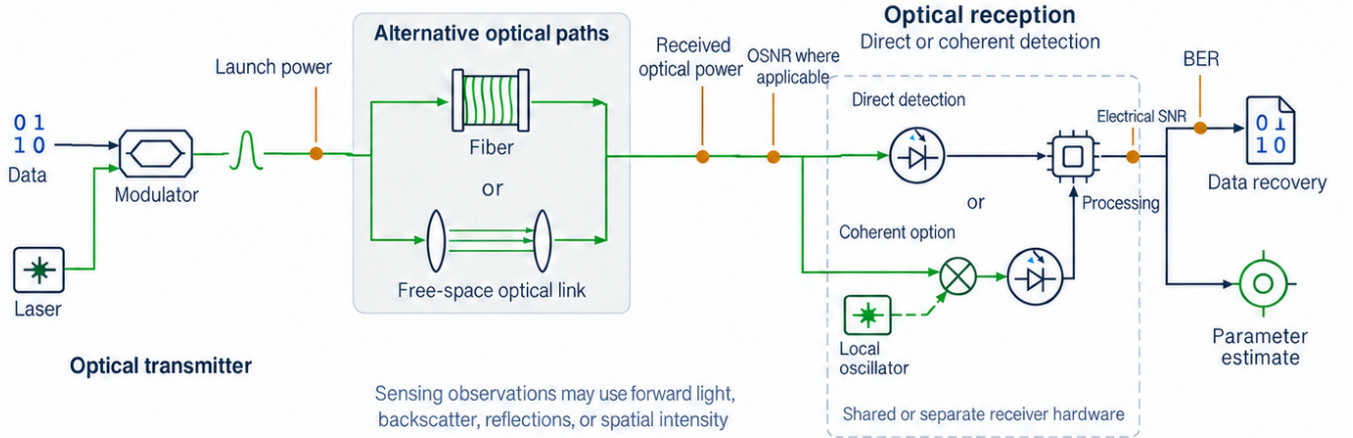
$$y(t) = \mathcal{R}_{\text{pd}}[h_{\theta} * x](t) + n(t), \quad x(t) \geq 0, \quad (1)$$

where t denotes time, $y(t)$ is received current in amperes, and $x(t)$ is transmitted optical power in watts. Photodetector responsivity $\mathcal{R}_{\text{pd}}$ converts optical power to current and has units of amperes per watt. The intensity impulse response h_{θ} depends on the physical parameters θ to be sensed, and $*$ denotes convolution. The additive term $n(t)$ represents electrical noise at the detector output.

This model relates optical power to a current observation. Coherent reception mixes the received field with a local oscillator and provides complex observations for amplitude and phase processing [1], [11].

The role of optics changes in photonics-enabled wireless systems. Optical tones can be mixed to generate a millimeter-wave or terahertz carrier at their difference frequency. The target is then interrogated by a radio-frequency (RF) signal, although optical sources and modulators determine its waveform. For example, a 262.1–268.3 GHz transmission is generated by mixing a swept optical source with a fixed optical tone [10]. Optical-source stability and wireless propagation therefore enter different stages of the same sensing and communication chain. Locating the interaction and receiver first

(a) Optical propagation and observation



(b) Photonics-enabled wireless propagation

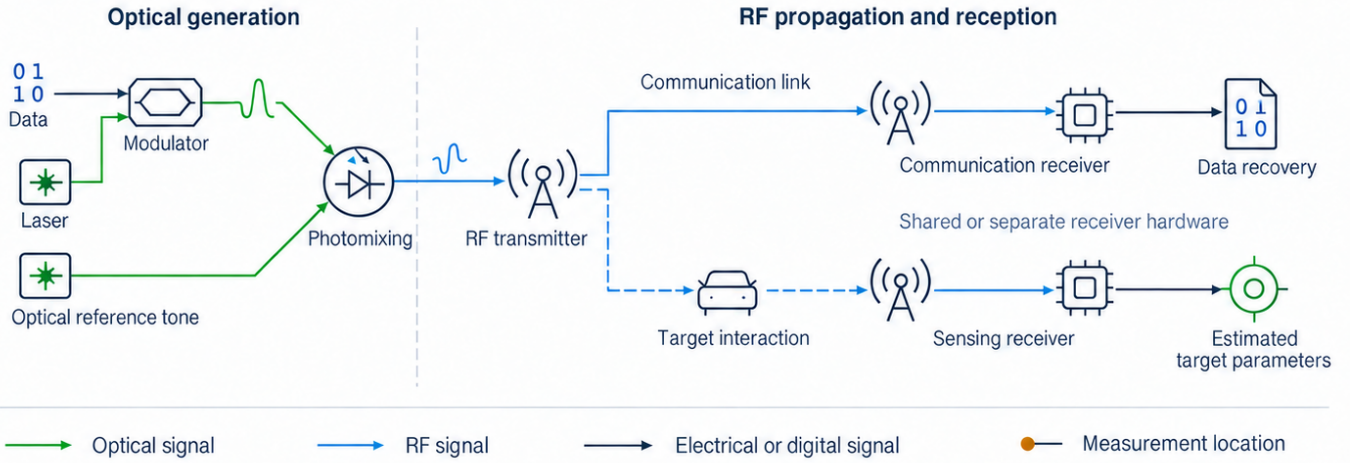


Fig. 2. Conceptual signal paths and measurement locations in O-ISAC. Panel (a) shows alternative fiber and free-space optical paths with direct or coherent reception. Sensing observations may use forward light, backscatter, reflections, or spatial intensity. Panel (b) separates optical generation from radio-frequency (RF) propagation and distinguishes communication and target-interaction paths. Receiver icons represent functional roles with shared or separate hardware. Markers identify launch and received optical power, optical signal-to-noise ratio (OSNR) before photodetection where applicable, electrical signal-to-noise ratio (SNR) after receiver processing, and bit error rate (BER) at data recovery.

makes it possible to identify the resources that each function can reuse.

B. Shared Resources and Optical Signal Constraints

Along the signal path, the shared resource may be a component, a transmission interval, or the waveform itself. Time sharing assigns communication and sensing to different slots. Frequency sharing assigns separate spectral portions, often with a guard band to limit interference. A joint waveform carries data while retaining structure that supports parameter estimation [11], [13]. Figure 3 illustrates these allocation patterns. Shared infrastructure can support any of them. The allocation determines which time and frequency resources are available to each function, while hardware constraints determine the waveform that can occupy them.

Optical hardware constrains the waveform that can realize this sharing. In an intensity-modulated link, a bipolar electrical

waveform must produce nonnegative optical power within the source's usable range. A clipped model with a direct-current (DC) bias makes this constraint explicit,

$$\begin{aligned} x(t) &= \min\{P_{\text{pk}}, \max[0, b + a u(t)]\}, \\ \mathbb{E}[x(t)] &\leq P_{\text{av}}, \end{aligned} \quad (2)$$

where $u(t)$ is a dimensionless real waveform, while a and b specify the modulation amplitude and bias in optical-power units. The peak and average power limits are P_{pk} and P_{av} , with $\mathbb{E}[x(t)]$ denoting mean optical power. Increasing the bias reduces clipping at zero but consumes average power and leaves less headroom below the peak limit. Clipping distorts the waveform used for data recovery and parameter estimation, so bias and modulation amplitude must be chosen together [11].

The power constraint applies to optical intensity. Complex coherent samples and the bipolar RF waveform after photonic

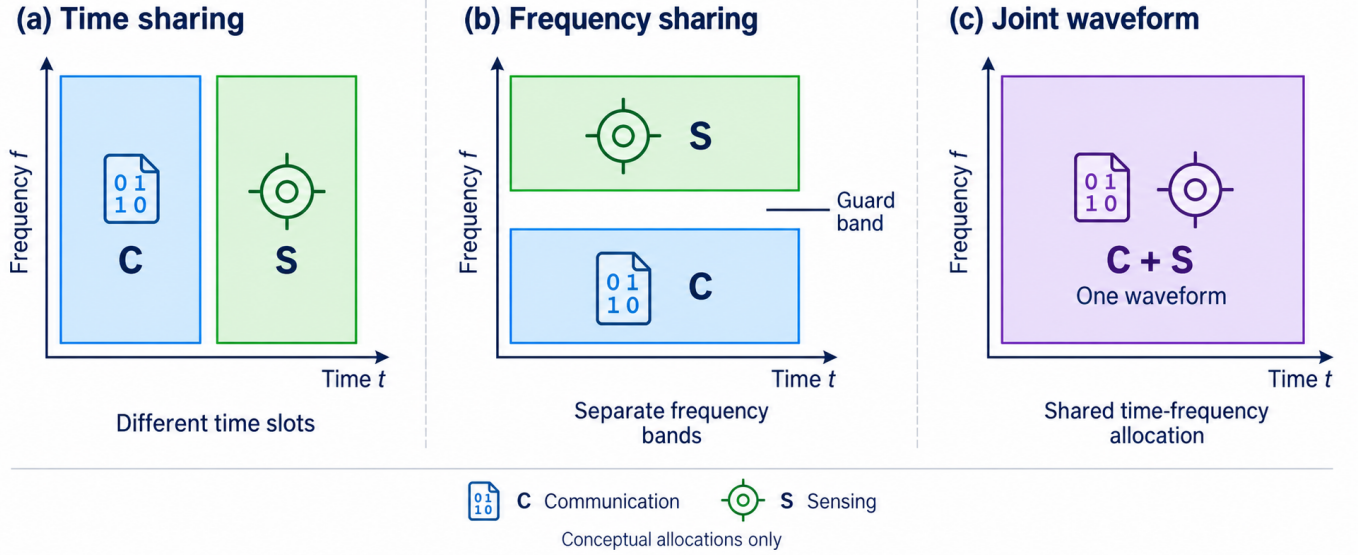


Fig. 3. Conceptual time-frequency allocations for communication and sensing. Panel (a) assigns different time slots to the two functions. Panel (b) assigns separate frequency bands, with a guard band shown in this example. Panel (c) uses one waveform within a shared allocation. The axes t and f denote time and frequency, and C and S denote communication and sensing. Icons identify the functions, while rectangles indicate allocated resources. Equal panel dimensions serve the illustration and do not assert equal bandwidth or performance.

conversion have different signal representations. In particular, nonnegative transmitted optical power does not require each electrical sample to be nonnegative.

An optical orthogonal frequency-division multiplexing (OFDM) simulation illustrates the bias choice. It uses 256 subcarriers spaced by 3.9 MHz and 32-symbol frames in a 200 m free-space optical channel model [11]. The nominal subcarrier grid spans $256 \times 3.9 = 998.4$ MHz, approximately 1 GHz. In this model, insufficient bias increases communication bit error rate through clipping, while excessive bias assigns more power to a component that carries no data. Range estimation is more tolerant of the clipping in the simulated settings, so its root mean square error responds differently to the same bias choice.

Sharing can also connect the functions through receiver information or spatial waveform structure. In a fiber experiment, a frequency-response estimate from the communication receiver is used to pre-compensate a sensing probe [1]. In a visible-light system, averaging the received pattern response supplies spatial information, while high-pass filtering recovers the temporal data modulation [2]. These mechanisms explain how the functions interact. Evaluating their effect requires measures that retain the meaning of each output.

C. Communication and Sensing Measures

The effect of a shared setting depends on what each performance measure counts. Communication rate describes how information is counted over time. For a frame carrying N_{payload} useful bits over a duration T_{frame} in seconds, net rate in bits per second is $R_{\text{net}} = N_{\text{payload}}/T_{\text{frame}}$. Symbol rate instead counts transmitted symbols, and its conversion to bit rate depends on modulation and overhead. Bit error rate (BER) measures the fraction of recovered bits that are incorrect at the stated processing stage. Delivered throughput also depends on decoding success and retransmissions. A rate estimated from

received signal quality describes a supported transmission rate under the assumed modulation and coding [2], [20], [21].

Sensing metrics distinguish target separation from parameter estimation. For two-way ranging in air, delay τ determines range r , while an ideal swept waveform gives the familiar nominal resolution

$$r = \frac{c\tau}{2}, \quad \Delta r \simeq \frac{c}{2B_s}, \quad (3)$$

where c is the propagation speed in air and B_s is the sensing sweep bandwidth in hertz. Delay is measured in seconds and range in metres. The resolution expression assumes a calibrated, bandwidth-limited delay response. Windowing, processing, and target conditions influence practical separation. A fiber delay model requires the appropriate propagation speed in the medium. Figure 4 shows this physical scale using the 6.2 GHz sweep reported in a photonic-THz experiment [10].

Estimation error answers how closely an estimated parameter matches its reference value. For N range estimates, the root mean square error (RMSE) is

$$\text{RMSE}_r = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{r}_i - r_i)^2}. \quad (4)$$

Here, $\hat{r}_i$ is the estimated range and r_i is its reference value for observation i , with both expressed in the same units. A maximum absolute error describes a different statistic. The same 6.2 GHz experiment reports a maximum ranging error of 0.55 cm for static targets at 20–45 cm [10]. This smaller error is consistent with estimating one target more precisely than the nominal separation scale for two targets. Practical resolution is established through a two-target measurement. A theoretical estimation bound instead describes the limit implied by its observation and noise model, which must be specified when comparing it with measured error [11].

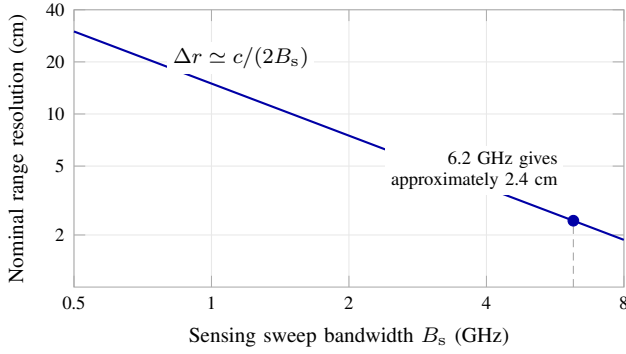


Fig. 4. Calculated nominal range resolution for an ideal two-way, bandwidth-limited measurement in air. The curve uses $\Delta r = c/(2B_s)$. The marked 6.2 GHz sweep corresponds to approximately 2.4 cm, as reported for the waveform in [10]. The calculation describes the nominal separation scale for two targets.

The sensing task also determines the meaning of a spatial dimension. Fiber spatial resolution describes localization along the cable, while gauge length specifies the separation over which differential phase is evaluated. The fiber example in Table II reports both quantities for this reason [1]. Visible-light localization likewise separates lateral and depth performance. Its ability to distinguish displacements along either direction is separate from the rate at which a moving receiver's location is updated [2].

Signal quality must likewise be tied to a point in the receiver chain. The optical signal-to-noise ratio (OSNR) compares optical signal and noise power in a stated reference bandwidth before photodetection. Electrical signal-to-noise ratio (SNR) depends on the detector, noise model, and electrical processing bandwidth. Launch power and received power similarly refer to different locations in Fig. 2. These distinctions explain why equal numbers in decibels need not represent equal receiver conditions [18], [21]. Once these meanings and measurement points are fixed, communication and sensing outcomes can be related to the setting that produced them.

D. Joint Operating Points and Performance Interpretation

A joint operating point links a shared design setting to both communication and sensing outcomes under a common, identified condition set. The results may come from one acquisition or from tests that explicitly retain the conditions needed for the claimed relationship. A claim of simultaneous operation additionally requires evidence that both functions operate together. Shared hardware alone establishes where integration occurs, while the performance relation depends on this connection between outcomes and settings.

Table II illustrates the meanings of reported measures and their evaluation conditions in three systems. Its rows collect results from the respective studies, including separate tests, and should be read as metric examples rather than single jointly measured operating points. The distinction preserves the contribution of each experiment while identifying which values can be interpreted together.

The clearest performance link is a controlled change in a shared factor. In the fiber example, increasing sensing-

probe power at fixed communication launch power raises sensing SNR while increasing the BER of both data subcarriers [1]. This is a within-study power tradeoff. The total launch power is not held fixed in this sweep. The pre-compensation gain listed in the table comes from a separate comparison using the communication receiver's response estimate. The two comparisons reveal resource competition and a benefit of cooperation under their respective conditions.

The other examples show why simultaneous signal generation and paired performance measurements require separate interpretation. The THz waveform carries data and a sensing sweep together, but communication and ranging are evaluated with different receiver configurations [10]. The visible-light study also demonstrates simultaneous data reception and dynamic localization, while the table's rate average and spatial resolving tests describe distinct evaluations [2]. The common waveform supports integration in both cases, and each numerical relationship retains its own test configuration.

Once both outcomes can be evaluated for a shared setting, joint design can express the choice as a constrained problem. An illustrative sensing-oriented formulation is

$$\begin{aligned} & \min_{\mathbf{u} \in \mathcal{U}} \text{RMSE}_s(\mathbf{u}) \\ & \text{subject to } R_{\text{net}}(\mathbf{u}) \geq R_0, \\ & \text{BER}(\mathbf{u}) \leq \epsilon_b, \end{aligned} \quad (5)$$

where RMSE_s is the root mean square error in the selected sensing quantity. The vector $\mathbf{u}$ contains the shared design variables, and $\mathcal{U}$ imposes power, bandwidth, and hardware constraints. The service requirements R_0 and ϵ_b fix the communication performance that must be retained. This formulation illustrates a service constraint, without assuming that every cited study solves this problem. A communication-oriented formulation instead maximizes a communication objective subject to a sensing requirement. A Pareto analysis examines settings for which improving one selected objective requires worsening the other within the feasible set [11], [13]. Establishing that frontier requires evaluating the feasible alternatives. A collection of separately reported best values does not establish it.

Interpreting results across studies adds physical and measurement conditions to this design view. Numerical comparisons require independent studies with compatible tasks, metric definitions, and measurement points. Their operating conditions and baselines must support the particular comparison, and a joint claim requires both outcomes to be linked to the shared setting. For an integration-gain comparison, a separate-function baseline must preserve the relevant resource budget and task conditions. Otherwise, a change in available power or measurement time can be mistaken for a gain due to sharing. When magnitudes are not directly comparable, a recurring mechanism can still explain a design choice, with each result retained in its own context.

The same condition set also determines what further evaluation can establish. Simulation predicts behavior under a model, while laboratory and field tests probe the system under their respective physical conditions. Section VI examines that validation evidence, and Section VIII derives priorities for

TABLE II
REPRESENTATIVE O-ISAC CONFIGURATIONS AND THE PHYSICAL MEANING OF THEIR REPORTED PERFORMANCE MEASURES

System / ref.	Signal and test configuration	Communication measure	Sensing measure and setting	Technical distinction
Fiber [1]	40 km fiber. Two 30 GBaud 16-QAM subcarriers and a 500 MHz LFM probe. A 200 Hz vibration is applied to a 1 m fiber segment.	Reported 2×120 Gbit/s during concurrent sensing. The probe-power sweep holds communication launch power at 0 dBm.	Reported 1 m nominal spatial resolution. Differential phase uses a 10 m gauge length. Probe pre-compensation improves vibration-spectrum SNR by 2.4 dB.	Span, spatial resolution, and gauge length have different roles. The 2.4 dB gain belongs to the pre-compensation comparison.
Visible light [2]	225 mW white laser diode and a 1 GHz receiver. Structured illumination covers 0.6×0.6 m at 2 m distance.	Estimated data rate averages 3.35 Gbit/s across patterns, with a 3.16–3.52 Gbit/s range.	Separate tests resolve 1 mm lateral and approximately 4 cm depth displacements. Dynamic localization operates at 39 frames/s.	The pattern-averaged rate and the static and dynamic localization values come from distinct tests.
Photonic THz [10]	262.1–268.3 GHz joint ASK/FMCW signal. The 6.2 GHz sweep lasts 400 ns per chirp. Communication is evaluated over 1 m.	15 Gbit/s ASK with measured BER 2.97×10^{-3} . The source reports approximately 36.2 GHz total occupied bandwidth.	Nominal range resolution is approximately 2.4 cm. Maximum ranging error is 0.55 cm for static targets at 20–45 cm, using a separate receiver configuration.	Sweep bandwidth differs from occupied bandwidth. Ranging error differs from resolution, and target distance differs from communication distance.

Values retain their source definitions and evaluation settings. Nominal resolution is distinguished from measured error. Values within a row can come from separate tests and do not by themselves define a joint operating point. Rates estimated from signal quality are distinct from delivered throughput. QAM denotes quadrature amplitude modulation and LFM denotes linear frequency modulation. ASK denotes amplitude-shift keying and FMCW denotes frequency-modulated continuous wave. BER denotes bit error rate and SNR denotes signal-to-noise ratio.

paired tests under disturbances and against separate-function baselines.

The study-selection method follows in Section III, while Sections IV and V use these foundations to examine architectures and joint-design results.

III. REVIEW PROCESS AND EVIDENCE BASE

The technical synthesis connects reported results to the physical setting and shared resources described in Section II. This section explains how we selected studies, extracted their results, and assessed their support for comparison. Review reporting follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement and its search extension, PRISMA-S [22], [23].

A. Search, Eligibility, and Study Reconciliation

We searched Scopus, IEEE Xplore, and ScienceDirect, together with SpringerLink, Wiley Online Library, and Taylor & Francis Online. The strategies covered O-ISAC phrases and combinations of joint sensing–communication terms with optical or photonic platform terms. All final searches were executed on 22 June 2026. Electronic Supplement S-Search provides the source-specific strategies and export records.

The publication window extended from 1 January 2020 through 22 June 2026, using the date when a report first became publicly available. Eligible reports were peer-reviewed journal articles or full conference papers with sufficient technical content in English. They addressed communication and sensing within a shared optical or photonic system and provided assessable evidence for at least one survey domain. Eligibility and support for a joint quantitative comparison were assessed separately. Contextual publications supported field positioning outside the primary study denominator.

Selection proceeded from deduplicated records through title and abstract screening to full-text assessment. Figure 5 shows the resulting record, report, and study counts. We reconciled reports describing the same underlying study before synthesis. This step mapped 227 eligible reports to 206 studies, with 21 companion reports retaining their source links. Electronic Supplement ST-01 provides this report-to-study mapping, and S-Exclusions records the eligibility decisions.

The review used an investigator-supervised workflow with assistance from artificial intelligence (AI) tools. These tools supported record preparation, extraction, and proposed adjudication. The investigator authorized specified decision packages, and rule-based checks verified source links and counting units. Routine independent duplicate human review and third-reviewer arbitration were not performed. Electronic Supplement S-Protocol documents the executed workflow and its departures from the original plan.

The review was retrospectively registered on the Open Science Framework on 12 February 2026 under record 7f6wb. That record describes an earlier review state. Dated amendments in S-Protocol explain the subsequent search scope and report reconciliation, together with changes in review conduct and synthesis. The final corpus reported here follows that amended workflow.

B. Extraction and Units of Analysis

A study can contribute several results obtained under different conditions. We therefore retained the report and source location for each extracted item. Metric records preserved the reported quantity and unit, the measurement point, and the test configuration. The configuration included operating conditions, baselines, and validation settings where supplied by the source. Electronic Supplement S-Data Dictionary defines the extraction fields.

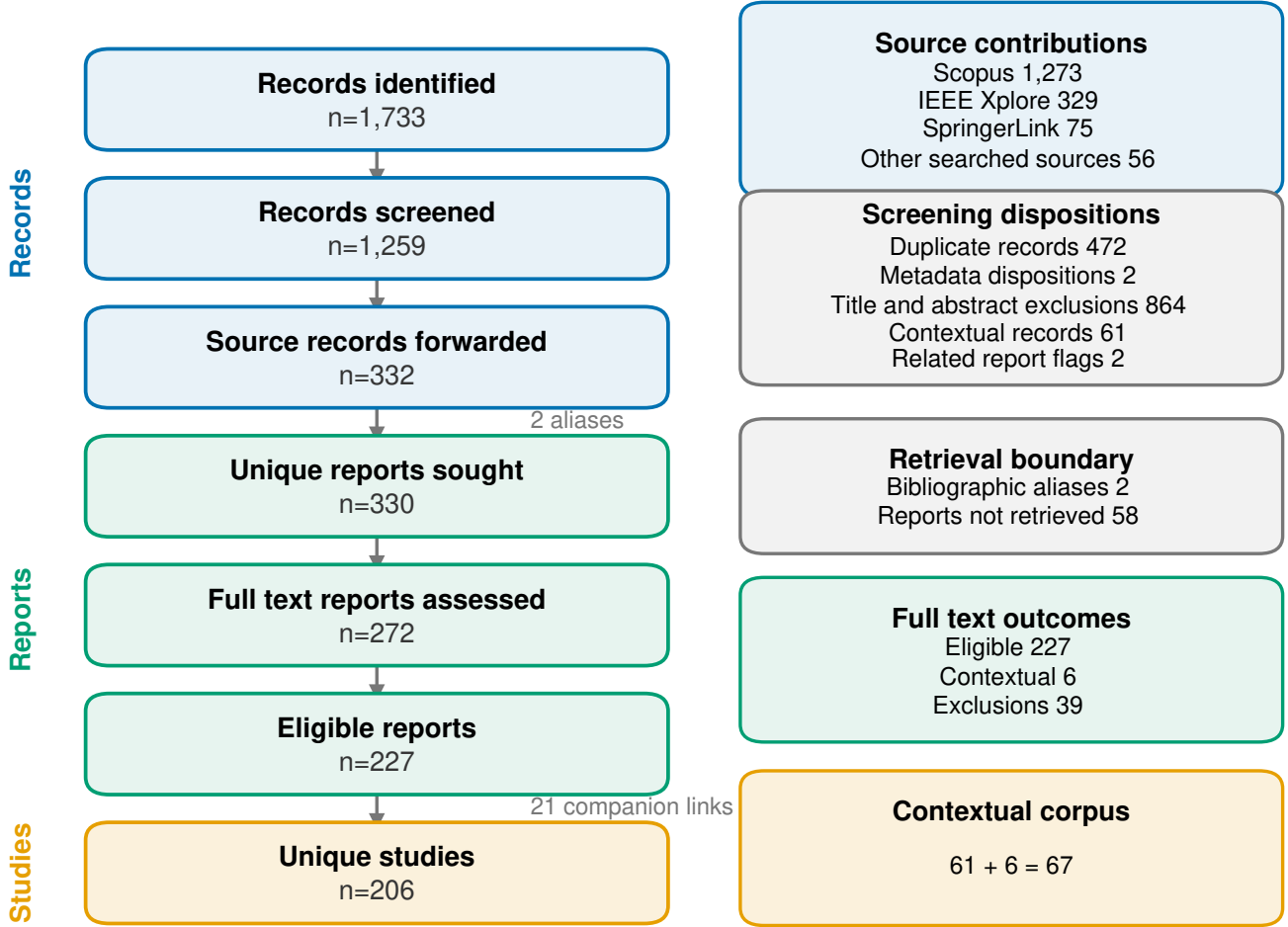


Fig. 5. Selection and reconciliation of the O-ISAC evidence base. Six sources yielded 1,733 records, of which 1,259 entered title and abstract screening. The retrieval route contained 330 unique reports after two bibliographic aliases were consolidated. Full-text assessment covered 272 reports and produced 227 eligible reports, mapped to 206 studies. The separate contextual corpus contains 67 records for field positioning.

Communication and sensing measurements remained separate unless the source linked them to a shared setting. Relationship records captured the design factors or objectives discussed in the source, together with their constraints and reported effects. These records support the mechanism-based synthesis, while Section II-D establishes the conditions for interpreting a numerical joint operating point. Table III distinguishes these analysis units.

Extraction retained numerical values stated by the source. Graphical results supported interpretation in their reported form, without creating additional numerical measurements. Source conflicts were handled at the individual-claim level, so supported findings from the same study remained available. The complete register contains 8,306 records, including 31 contextual items and 72 items with source conflicts outside the primary synthesis. Electronic Supplement S-Evidence provides the primary synthesis records and their source links. S-Protocol describes the rules for handling contextual items and source conflicts.

C. Technical Appraisal and Synthesis

We assessed all included studies using the Technical Quality Assessment Framework (TQAF) developed for this review. Its deterministic rules score eight technical and reporting dimensions from 0 to 3. Overall evidence contribution is recorded separately. TQAF is a review-specific descriptive appraisal whose scoring has not been independently validated. It characterizes support for interpretation and reuse, rather than providing a conventional risk-of-bias assessment.

Figure 6 separates the appraisal dimensions by their role in reading the evidence. Metric clarity and reporting completeness help establish what a result means in its original setting. Validation and reproducibility address how that result was tested and can be reconstructed. Benchmark readiness and comparison admissibility address its use beyond the original experiment. Electronic Supplements S-Protocol and S-Appraisal provide the scoring rules and study-level records, respectively.

The synthesis grouped studies by technical theme across seven survey domains. Within each theme, we examined re-

TABLE III
UNITS OF ANALYSIS AND THEIR ROLES IN THE TECHNICAL SYNTHESIS

Analysis unit	Count	Meaning of one entry	Use in the survey
Included study	206	An underlying study after reconciliation of its eligible reports.	Study-level platform, integration, and validation profiles.
Primary coding record	8,203	An extracted evidence item, metric record, or tradeoff record retained for primary synthesis.	Source-linked technical analysis, with each record retaining its own unit and conditions.
Metric record	4,779	A reported measure with its definition and evaluation setting.	Communication, sensing, and implementation performance in Section V.
Substantive relationship	402	A source-described relation among design factors, objectives, or outcomes.	Mechanisms and condition-dependent tradeoffs in Section V.
Synthesis group	115	A set of studies grouped for thematic synthesis.	111 theme-specific groups support synthesis. Four broad classification groups retain coverage information.

The counts refer to different analysis levels. Metric and relationship records contribute to the primary coding total, while a study may contribute to several synthesis groups. The primary total comprises 3,020 evidence items, 4,779 metric records, and 404 tradeoff records. Of the latter, 402 describe relationships and two document the absence of a reported tradeoff.

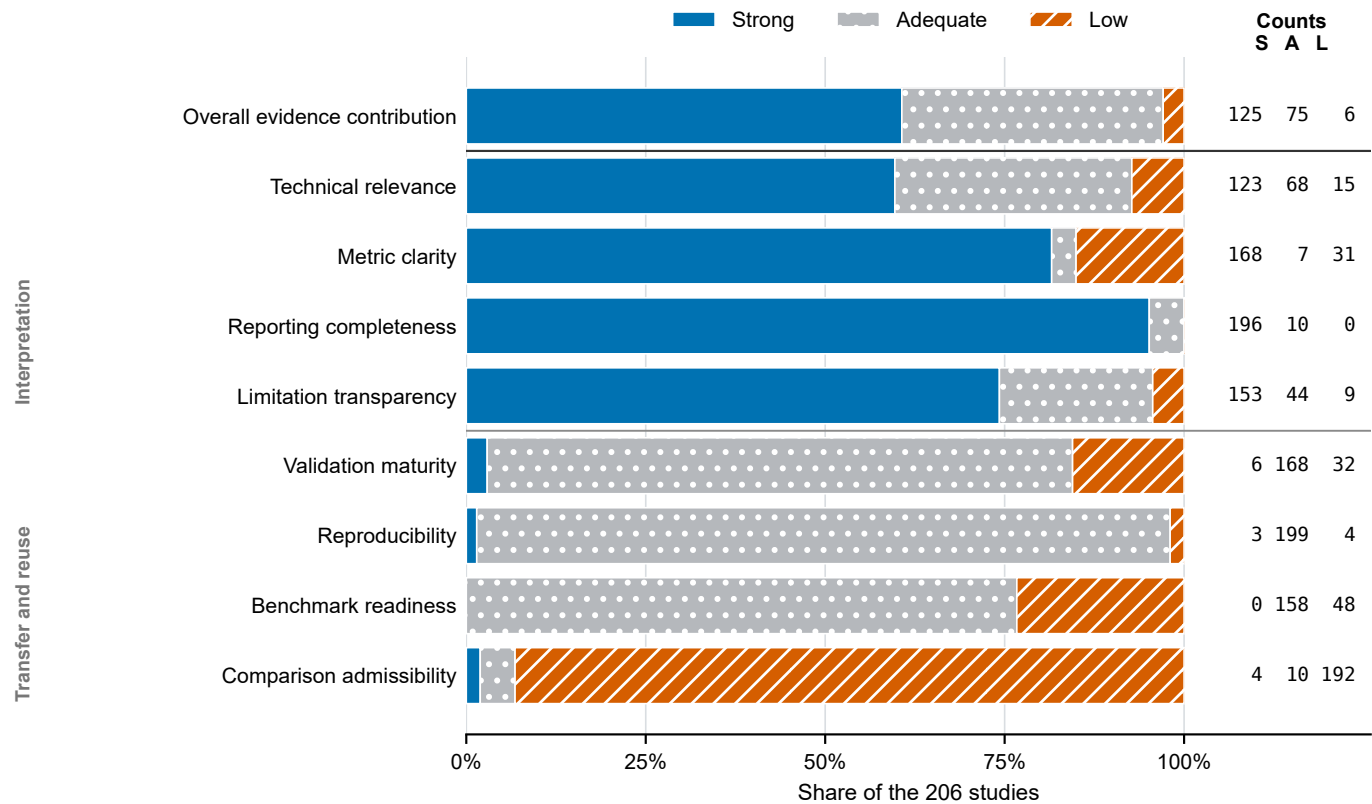


Fig. 6. Technical appraisal of the 206 included studies under the review-specific TQAF rules. Each row groups all studies by strong, adequate, or low support. The interpretation dimensions concern the meaning and reporting of results in their source setting. Transfer and reuse dimensions concern validation and reconstruction, together with benchmark and comparison support. Overall evidence contribution is displayed separately.

curing mechanisms and reported performance relations while preserving their physical and measurement conditions. Thematic groups supported synthesis conclusions, while broad classification groups recorded coverage across mixed topics. Electronic Supplement S-Bodies provides group membership and the review-specific appraisal rules.

We used structured narrative synthesis to connect these heterogeneous results. Numerical examples retain their original tasks, units, and operating conditions. The review did not fit a pooled effect model or perform formal publication-bias

and sensitivity analyses. The resulting conclusions concern supported mechanisms and bounded performance relations.

Section IV first examines the physical role of optics and the resources shared by communication and sensing. Those distinctions provide the context for the performance relations in Section V and their validation in Section VI.

IV. OPTICAL PLATFORMS AND INTEGRATION ARCHITECTURES

In O-ISAC, the physical role of optics determines both where information propagates and where the environment is

TABLE IV
CANONICAL PHYSICAL ORGANIZATION OF THE CORPUS OF 206 STUDIES

Platform family	Studies, n (%)
Photonic terahertz	69 (33.5%)
Fiber	56 (27.2%)
Visible light communication and LiFi	38 (18.4%)
Free space optical	31 (15.0%)
Hybrid optical	9 (4.4%)
Other optical	3 (1.5%)
Total	206 (100.0%)

Note. Each study is counted once under its dominant platform family. Media and integration mechanisms can overlap within the represented systems. The counts summarize how the reviewed evidence is distributed across the six platform families.

observed. Optics may carry the user link, illuminate a target, distribute a wireless carrier, or provide a coherent reference, and each role exposes a different set of impairments. The integration architecture then reveals which component, signal, or resource is shared by communication and sensing. We therefore examine the evidence along the axes of physical platform and coupling mechanism.

Table IV reports the mutually exclusive family counts and corpus shares. Photonic terahertz and fiber together account for 125 studies, representing 60.7% of the corpus.

The family profiles below provide the corresponding physical paths, observables, coupling mechanisms, active constraints, transfer conditions, and evidence anchors. Each study belongs to one platform family, while its coupling mechanisms may overlap. Transmitters supplied through fiber enter the terahertz family when photonic generation and conversion create the wireless carrier directed at the target. Fiber remains the organizing platform when it transports signals to downstream radio frequency identification (RFID), radio, or radar sensing. Hybrid membership begins when interaction between physical segments shapes the result. This organization locates the target path and guides the synthesis that follows.

A. Physical Platform Families

1) *Photonic Terahertz Systems*: Photonic terahertz is the largest family, comprising 69 of the 206 studies (33.5%). The optical chain generates, distributes, or processes a coherent carrier before conversion to a wireless terahertz or millimeter wave target path. It yields delay, range, Doppler, angle, imaging, and channel or target response observables. Communication and sensing are coupled through hardware, references, waveforms, beams, or resources [9], [10], [20]. This upstream sharing ties both functions to the same optical conditions. Laser coherence and linewidth govern reference stability, while component bandwidth, conversion fidelity, and efficiency shape the bandwidth, phase structure, and power delivered to the antenna [24], [25]. At the radio interface, antenna aperture, gain, and alignment govern launch and collection, molecular absorption shapes propagation, and target scattering forms the returning echo in radar configurations [26], [27].

Where the functions meet varies by architecture, with some systems dividing a multicarrier frame between sensing and

communication and others placing the joint decision in a phased array controlled through photonics [9], [20]. Wavelength tuning and multibeam reception move coordination to the source or the receiving front end [28], [29]. At the waveform level, one multichannel design combines linear frequency modulation (LFM) with communication symbols across coherent carriers [24]. Other designs add source agility through frequency hopping or organize the signal over delay and Doppler through orthogonal multiplexing [27], [30]. Information can also close the loop, as burst SNR estimates guide both probabilistic shaping and ranging, while a simulated radar waveform carries communication data [26], [31]. Together, these designs show how the joint decision can move from hardware to waveform and to the information used by both tasks.

The location of that decision also determines how the measurement is formed. A returned echo may be remodulated onto light and carried back over fiber for centralized processing, where photonic dechirping expresses target delay at a lower frequency [27], [32]. Along this path, phase noise associated with laser linewidth degrades the recovered signal, while delay mismatch between optical and electrical paths shifts the ranging observable [32]. Laboratory experiments reveal waveform, conversion, and beamforming behavior under controlled conditions [9], [24], whereas a 270 m outdoor link adds wireless propagation and operation of the complete chain [20]. Comparison across photonic terahertz systems depends on aligned optical and wireless stages, aperture and beam conditions, receiver processing, and validation setting.

2) *Fiber O-ISAC*: Fiber comprises 56 studies (27.2%) and follows two physical paths. The guided link may form the sensing medium, with disturbances observed through backscatter, carrier phase, modal coupling, or an inserted sensor [33], [34], [35], [36]. Fiber may instead transport signals to a downstream RFID, radio, or radar segment, so the target interaction occurs after optical distribution [37], [38], [39], [40], [41]. The corresponding observations include distributed events, vibration, strain, and environmental change in the guided path, together with downstream tag state, range, localization, and imaging [33], [37], [38]. The measurement boundary therefore lies either within the guided path or across the complete optical and wireless chain.

Coupling recurs through shared hardware, wavelengths, waveforms, links, or resource decisions. Integration can reuse the communication signal itself [34], [35], [42], [43], [44] or place distinct data and sensing signals on shared infrastructure [3], [33], [36], [45], [46]. The first form ties sensing to the symbols, carrier, phase, or supported modes, whereas the second ties both functions to the same fiber and receiver conditions. Guided sensing is constrained by dynamic range, leakage, reflection, nonlinearity, reference stability, topology, and traffic, whereas transported sensing additionally depends on conversion, timing, remote calibration, and wireless geometry. Evidence comes from laboratory links, live access networks, and a deployed submarine route. For fiber evidence, comparison depends on the interrogation or transport role, topology, launch and receiver conditions, conversion and clocking, target path, traffic, and validation setting [33], [37],

[38], [41], [42], [43], [46].

3) *Visible Light Communication and LiFi Systems*: Visible light communication and LiFi comprise 38 studies (18.4%) and place communication and sensing within the same illuminated scene. The grouping preserves their distinct meanings at the link and networking levels. Direct or reflected room and device paths through emitters, cameras, and photodiodes support position, pose, distance, imaging, gesture, recognition, and physiological sensing [2], [47], [48]. Cameras and photodiodes expose different balances between spatial detail and reception speed, while nonnegativity, dimming, flicker, source linearity, ambient light, field of view, and exposure constrain what can be communicated and observed [47], [49], [50], [51]. Coupling recurs through hardware, waveform, geometry, resource decisions, and the application loop. It may arise through a shared observation [48], [52], through geometry and resource decisions [53], [54], [55], [56], [57], or through integrated emitters and detectors [58], [59], [60].

Illumination is the common comparison boundary across these designs. Geometry and multipath shape localization, reflected observations support gesture and recognition, and learned models extend the same optical field to health sensing tasks [47], [51], [55], [56], [61], [62]. Analytical and simulated studies isolate geometry and resource choices, whereas prototypes and field experiments add lighting, receiver, and control conditions [47], [52], [53], [55], [57], [59]. Visible light comparisons depend on matched task definitions, layout, orientation, illumination, detector, processing, and validation conditions.

4) *Free Space Optical Systems*: Free space optical systems comprise 31 studies (15.0%) and use a directional link, optionally combined with target illumination and return, over direct or reflected aperture and beam paths. These paths support ranging, localization, imaging, tracking, and atmospheric or channel observation [63], [64], [65]. Geometry is therefore part of both the link and the measurement. Coupling recurs through the link, waveform, aperture or beam, and spatial or temporal resources, while pointing, divergence, aperture, reflectivity, loss, turbulence, and motion constrain the resulting evidence. Direct, reflected, scanned, and steered paths illustrate these relations [63], [66], [67], [68]. Retroreflection and multicarrier reception connect returned power to range and position [64], [65]. Structured waveforms and steering apertures place coupling in the signal or beam [69], [70], [71], [72], while atmospheric estimates can guide beam shaping or link selection [8], [73].

Across these forms, beam geometry determines both spatial selectivity and the power collected at the receiver. Range, angle, and image features therefore remain tied to calibration, aperture, pointing, and motion [63], [64], [65], [66], [67]. Controlled links reveal waveform and estimator behavior, whereas operational ranging adds atmospheric variation and continuous field observations [8], [65]. Comparison across free space optical systems depends on matched geometry, calibration, target properties, weather, motion, and validation conditions. Hybrid systems extend that comparison across interfaces between physical segments.

5) *Hybrid Optical Systems*: Hybrid optical systems comprise nine studies (4.4%) in which consequential interaction occurs among optical, fiber, free space, or radio segments and the perturbation and observation may straddle a conversion boundary. Their observables include range, vibration, and target or network state within individual segments, while coupling arises through a central reference and through links, resources, control, or applications shared across segments. Conversion loss, segment power, synchronization, calibration, and timing are therefore active constraints [74], [75], [76]. Hybrid comparisons trace every segment, conversion, clock, and measurement plane together with the validation of both functions.

Unlike fiber used only for downstream transport, hybrid membership requires conversion or feedback across segments to shape the result. This occurs when interaction between fiber and wireless segments is consequential [74], [75], when a shared photonic source spans fiber, free space, and millimeter wave stages [77], or when sensing in one segment guides service in another [76], [78]. Other designs establish a common signal origin through shared time modes or a photonic waveform with frequency agility [79], [80].

6) *Other Optical Systems*: The three other optical studies (1.5%) use distinct mechanisms based on optical control or coherent carriers. These comprise a microwave field generated by a microcomb, a solar blind ultraviolet path, and a quantum optical path. Their observables are, respectively, a field pattern, particulate or smoke response, and phase [81], [82], [83]. Coupling is specific to specialized hardware, reference, and application choices, and each case is bounded by its own propagation, target, receiver, and measurement models. Each numerical result is therefore interpreted within the native mechanism and measurement model of its system. Together, the hybrid and other optical cases complete the physical map before the analysis turns to coupling mechanisms shared across families.

B. Integration Architectures and Resource Coupling

Across these physical families, integration architecture locates where communication and sensing meet along the target path. The coupling may enter through shared hardware, a common carrier or waveform, coordinated resources, a propagation link, or a service decision. Several mechanisms can coexist in one system. Each label identifies one coupling location while preserving the rest of the chain for separate interpretation.

Figure 7 locates seven recurring coupling mechanisms along a generic signal and service path. A study may occupy several locations, so the badges report overlapping coverage across the corpus.

Resource, hardware, and waveform sharing can form one connected design chain. Application coupling acts at the service level and carries a different engineering meaning. The following subsections trace these locations from physical reuse to coordinated resources and design.

1) *Shared Hardware, Carrier, and Link*: Hardware, carrier, and link sharing describe different physical relations. Reusing a component makes its bandwidth, coherence, dynamic range,

Coupling locations may overlap within one study

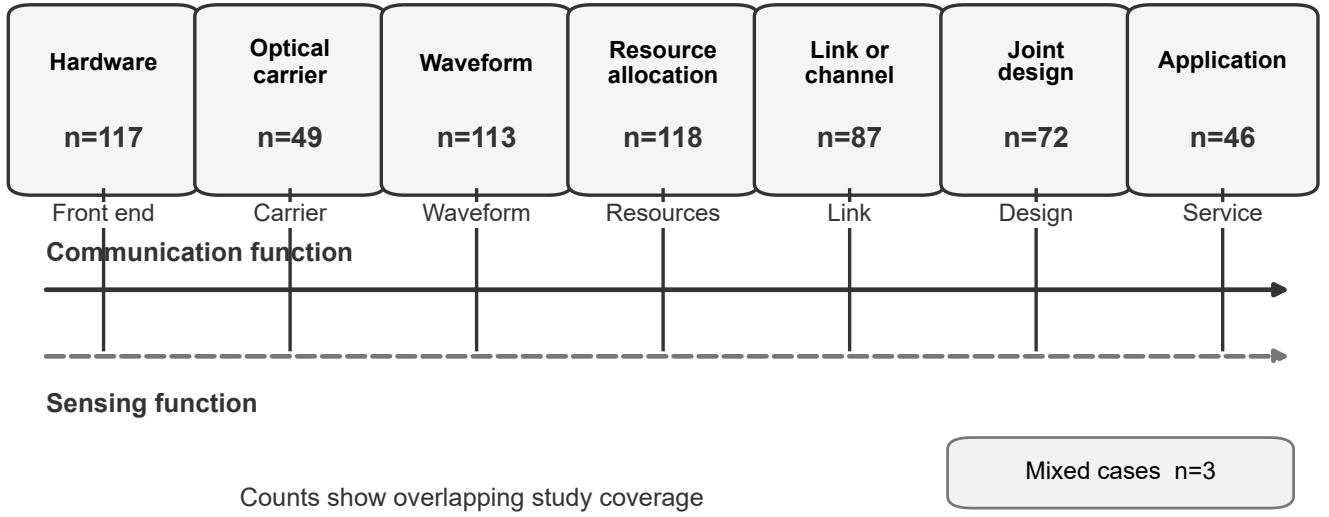


Fig. 7. Locations of communication and sensing coupling in the corpus of 206 studies. Equal cards identify hardware, optical carrier, waveform, resource, link, joint design, and application coupling along the system path. The badges report overlapping study coverage, with three mixed boundary cases shown separately. Card position identifies the typical coupling location, while each card retains equal visual weight.

or alignment conditions relevant to both functions. A common optical carrier adds a phase or wavelength relation, while a common link adds shared propagation and calibration conditions. Together, these relations show which design choice is common and which constraint follows it.

The physical path gives reuse its engineering meaning. In fiber, traffic and sensing share the guided structure, with a resonance shift, phase, or backscatter conveying the sensed change [3], [84]. A retroreflective free space path supports information delivery together with ranging or localization [63]. Arrays controlled through photonics instead share optical generation and beam control while the target path remains wireless [9]. Comparison therefore follows the component role, carrier relation, propagation path, and calibration.

2) *Shared Waveforms*: Waveform sharing begins when one signal structure contributes to both decoding and sensing. The shared element may be the complete waveform or a selected pilot, chirp, code, polarization state, or spatial pattern. This common design links payload efficiency, sensing observability, and receiver processing. Structural sharing records that relationship. Concurrent performance adds the requirement that both outcomes be observed at a matched operating point, a distinction carried into Section V.

The evidence illustrates several forms. Pilots inserted for channel estimation can also provide a sensing reference [85]. One photonic terahertz prototype generates a waveform whose frequency steps provide the radar dimension and whose amplitude carries data. Its experimental evidence is strongest for waveform generation and communication response [86]. A visible light design combines temporal data with spatially structured illumination and uses localization to adapt beam-

forming [2]. The shared structure is similar, while its physical and receiver implementation remains specific to the platform.

3) *Resource Allocation and Joint Design*: Resource coupling coordinates temporal, spectral, power, spatial, or processing degrees of freedom. An allocation can change payload opportunity, link margin, sensing observability, geometry, or latency. Joint design connects the two functions more directly by choosing a variable, objective, or constraint in view of both outcomes. The resource, allocation rule, and operating conditions therefore define the resulting evidence.

A visible light model for vehicle to everything communication makes exposure time a shared decision variable. Its coexistence result is analytical and numerical, while laboratory communication and sensing gains were obtained in separate configurations [49]. In one reported terahertz configuration, a rapidly swept photonic source and common waveform plan support simultaneous data transmission and ranging [10]. These cases distinguish a modeled optimum from a measured concurrent configuration. Section V therefore reads each relationship within its operating region and evidence form.

4) *Coupling at the Application Level and Boundary Cases*: Application coupling arises when sensing informs beam or link selection, trajectory planning, blockage response, or network control. The shared element is then the system action, even when the signal paths remain separate. A near infrared wearable combines communication and physiological sensing on a common material platform while retaining separate detector roles [60]. A hybrid last mile design places free space optical delivery and fiber Bragg grating monitoring within one system model [76]. Their coordination lies in hardware and service context.

Three studies cross temporal or physical segment boundaries. One performs localization and imaging before data delivery on a shared optical source and path with separate receivers [66]. Two use sensed free space optical channel state to adapt trajectory and resource allocation or cell placement and user association across optical backhaul and radio access [87], [88]. The mixed label preserves this path and keeps each result tied to the segment, function, and operating stage that produces it.

C. Interpretation Across Platforms

Read together, the platform and architecture views form one mechanism map. The platform establishes the physical transfer conditions, while the architecture identifies the shared choice. In fiber, waveform sharing inherits guided propagation and reflections, together with launch and receiver conditions [42], [43]. In free space optical systems, the same mechanism is shaped by aperture, geometry, and alignment, whereas terahertz systems assisted by photonics also introduce conversion from optics to radio [9], [63]. Application coupling can instead coordinate separate optical and radio paths through a common service decision [87].

Across these mechanisms, sharing concentrates design around a common component, reference, or resource. That choice can couple the functions through power balance and interference, while synchronization and receiver dynamic range constrain their joint operation [3], [75]. Section V therefore examines which communication and sensing outcomes change together at a matched operating point and which conditions explain that relationship.

V. PERFORMANCE METRICS AND JOINT DESIGN TRADEOFFS

After locating a shared choice, the analysis turns to the reported communication and sensing outcomes. A reported value gains engineering meaning from its task, its observation point in the signal chain, and its operating and validation conditions. The comparison profile connects these elements to the performance evidence.

The primary synthesis contains 4,779 metric records. Each record is an extraction unit that retains the reported quantity together with the definition and source conditions available for it. The denominator therefore represents reported quantities in their original contexts. Data rate, bit error rate (BER), and range resolution have the widest study coverage, appearing in 85, 60, and 50 studies, respectively. Since one study may report several families, these totals provide overlapping views of the evidence.

Beyond these familiar measures, the evidence extends from link and waveform behavior to outcomes observed at the device, estimator, application, and implementation levels. This breadth makes semantic alignment the organizing principle for the analysis that follows.

A. Applying the Comparison Profile to Individual Metrics

Applying the comparison profile defined in Section II, each metric row was coded by source definition, task, measurement plane, operating point, validation method, and value

provenance. Communication and sensing results form a joint operating point when the source links them to the same condition set. Table V summarizes domain coverage and the analytical role assigned to each record.

TABLE V
METRIC EVIDENCE PROFILE FOR 4,779 PRIMARY RECORDS

<i>Metric domains</i>		
Domain	Records, <i>n</i> (%)	Studies, <i>n</i> (%)
Sensing	1,816 (38.0%)	203 (98.5%)
Communication	1,328 (27.8%)	194 (94.2%)
Joint	870 (18.2%)	158 (76.7%)
Implementation	476 (10.0%)	64 (31.1%)
Smaller normalized domains	289 (6.0%)	33 (16.0%)
<i>Analytical roles</i>		
Role	Records, <i>n</i> (%)	Use in the survey
Bounded relation	118 (2.5%)	Conditional comparison across studies
Native context	4,661 (97.5%)	Mechanism, feasibility, and qualitative synthesis

Note. Domain rows and analytical roles each partition the 4,779 primary records, while study counts overlap. The complete metric register contains 4,861 rows. The additional rows comprise 31 contextual records and 51 records documenting source conflicts. Comparison follows the reported definition, task, measurement plane, conditions, and validation. Electronic Supplement S-Evidence provides the complete records and provenance, including quantities grouped under the broader other category.

The domain distribution shows where metric evidence is concentrated, while the analytical role determines how each record enters the survey. The following subsections explain how task, measurement plane, and operating conditions shape communication, sensing, joint, and implementation metrics.

B. Metrics Across Functions

1) *Communication Metrics:* A communication rate gains meaning from what the source counts. Symbol rate tracks signaling pace, while data rate reflects the payload and overhead included in that count. Throughput describes delivered service, capacity provides a model bound, and spectral efficiency relates useful data to its stated denominator. In a terahertz prototype enabled by photonics, net rates follow communication subcarrier allocation, payload occupancy, and coding overhead [20]. An embedded pilot orthogonal frequency division multiplexing (OFDM) simulation defines spectral efficiency as the data subcarrier fraction and reports cyclic prefix overhead separately [85].

Reliability shifts attention to the event and its location in the signal chain. BER measured before and after forward error correction describes two stages of the link, while packet loss and outage describe broader delivery boundaries. A fiber experiment over a long span reports receiver OSNR and BER before correction during concurrent distributed sensing [21]. A robotics study reports packet loss over distance and view angle, display delay over queue size, and formation control under dynamic motion as separate results [47]. The reported reliability value is therefore tied to its event, observation point, time scale, and system boundary.

The observation point also determines how signal quality and power are interpreted. OSNR relates signal power to noise in a stated optical reference bandwidth, while electrical SNR follows photodetection and digital SNR follows equalization. Launch and received power likewise refer to the transmitter and receiver ends of the link. A fiber polarization sensing experiment traces recovered vibration SNR across an OSNR sweep, connecting the sensing response to a stated optical condition [18].

Distance and latency complete the profile by locating the result in space and time. Fiber span and wireless separation describe link geometry, while target range and coverage radius describe the observed or served region. Latency may include propagation and buffering, signal acquisition and processing, or the full application loop. Stated endpoints, included stages, and operating conditions define the scope of each value. The complete profile connects each communication metric to the system behavior it represents.

2) *Sensing Metrics*: Sensing metrics first take meaning from the task and quantity they represent. Range and position estimates, motion measurements, images, and class decisions refer to distinct observations and references. Resolution describes separability along a stated dimension, while estimation error relates an estimate to a reference. Repeatability captures variation across trials, and a model bound supplies a conditional benchmark. These distinctions establish the comparison basis.

Representative studies show how this contract works. A fiber reflectometry system reports breakpoint localization error and range resolution separately for laboratory and simulated conditions [89]. A visible light localization experiment distinguishes lateral estimation error from the smallest demonstrated displacement and summarizes signed displacement with means and standard deviations across repeated trials [2]. An optical phased array simulation relates imaging root mean square error to electrical SNR under a defined scene, waveform, and beamforming constraint [90]. Each result retains its summary, geometry, and validation setting.

Detection and classification add dataset and decision context. Their metrics take meaning from class balance, decision thresholds, background conditions, and the relation between training and test data. CORE-Lens reports recognition accuracy and optical camera communication BER across stated class, data rate, and background settings, with runtime evaluated separately [91]. VehicleTalk evaluates link performance, distance estimation error, and vehicle status recognition under separate prototype conditions [48]. Their application meaning follows from the attached classes, geometries, and test protocols.

Physical quantity sensing follows the same logic by linking each measurand to a reference and calibration. Spatial support and acquisition time locate the observation, while the aggregation rule shapes its summary. The resulting profile connects the task, reference, geometry, processing, and validation setting to the reported sensing outcome. Together, these elements define the conditions for comparison.

3) *Joint and Implementation Metrics*: Joint metrics connect communication and sensing through a shared operating point

or resource rule. A matched tuple uses the same conditions for both functions. A shared resource can also link values that retain their own reported conditions. In an embedded pilot OFDM simulation, BER and mean absolute pilot error are reported at the same SNR with eight pilots, while spectral efficiency follows the corresponding data subcarrier fraction [85]. The pilot error measures deviation of known pilot symbols in the simulated channel. Together, the three values show how pilot allocation connects communication reliability, resource efficiency, and the reported sensing measure.

Optimization creates another form of joint evidence by selecting an operating point through shared design variables. In a simulated optical phased array system, precoding matrices and photodiode orientations are selected together. The design maximizes a sensing contrast metric under a communication quality threshold and emitter power constraints [90]. Geometry, constraint thresholds, and metric definitions define the scope of the solution.

Implementation metrics place these operating points within the platform and application chain. Receiver bandwidth, converter rate, memory, acquisition time, and latency show where processing burdens arise. An outdoor photonics assisted terahertz prototype evaluates receiver pipeline latency in hardware analysis and communication and ranging performance in a field experiment [20]. A smartphone camera prototype evaluates recognition and communication in the laboratory and benchmarks module latency, memory, and complete cycle timing on the device [91]. Together, these measures connect joint performance to the processing chain and locate the effects of shared design choices in system behavior.

C. How Communication and Sensing Performance Interact

Communication and sensing often share bandwidth, power, waveform components, hardware, and processing. A design change in any of these elements can move both functions to a new operating point. More pilot symbols may strengthen a sensing reference while leaving fewer resources for data. Wider bandwidth may support both a higher data rate and finer range resolution until a receiver or platform limit becomes active. The evidence is organized by the shared choice that produces each pattern.

The synthesis covers 402 relationships from 168 studies. Figure 8 separates record type, overlapping study coverage, and condition dependence across 11 recurring mechanism families. Each relationship remains tied to its waveform, geometry, target, observation point, and validation setting.

Resource and power mechanisms carry much of the coded evidence, while condition dependence remains common across the families. This pattern directs the discussion toward the shared mechanism and the operating condition that shapes each reported relation.

1) *Bandwidth, Spectrum, and Resource Allocation*: Bandwidth links the two functions when both draw on the same signal space. That space may be divided across subcarriers, wavelengths, guard intervals, observation time, or the usable response of a device. Embedded pilots show the basic pattern. Increasing pilot support improves the reported pilot deviation

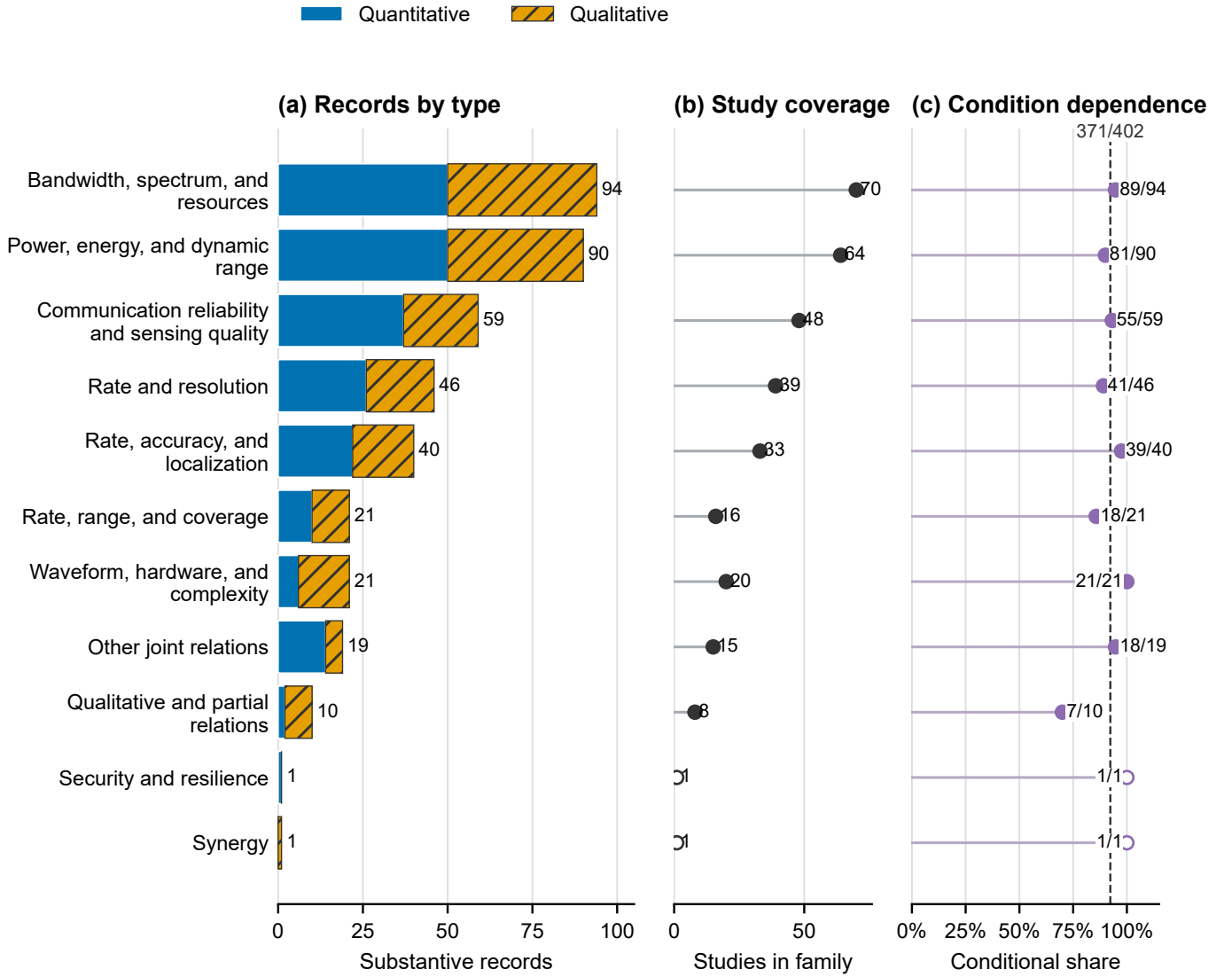


Fig. 8. Evidence profile for 402 substantive joint performance relationships from 168 studies. Panel (a) separates 218 quantitative and 184 qualitative records. Panel (b) reports overlapping study coverage across the mechanism families. Panel (c) reports the share of records whose interpretation depends on stated conditions, with 371 such relationships across the profile. Family study counts may overlap, and frequency describes corpus coverage.

proxy while changing BER and the data carrying fraction [85]. Protection intervals and chirp sweeps create related exchanges by improving sensing isolation or pulse compression within device bandwidth and subcarrier limits [45], [75]. In each case, the effect follows the resource being shared and the limit that becomes active.

Waveform structure determines how the shared signal space is used, with format and operating mode setting time and spectral sharing [10], [63], [66]. Within that structure, digital designs assign subbands, chirp parameters, power, and weighted resources [92], [93], [94], [95], [96]. Pilot strength, placement, and synchronization fields define reference support and communication margin, while receiver partitioning and ranging sequence redundancy set the usable signal region [20], [97], [98], [99], [100], [101]. A tunable laser study supplies qualitative generation context for the same design space [102]. Together, the allocation rule and receiver condition define the

operating point reported in each design.

Resource sharing also appears in time. Observation duration, camera exposure, and task scheduling connect sensing quality and cadence with communication goodput or execution time [49], [91], [103]. Time division, chirp repetition, coding traffic, security control, and embedded pulse count place both functions within a shared frame [48], [52], [104], [105], [106]. Other designs reserve spectral, temporal, spatial, polarization, or phase structure to manage interference, coverage, and receiver complexity [7], [29], [40], [107], [108], [109]. The observed outcome therefore depends on the frame period, exposure response, isolation rule, and remaining occupancy.

When both functions occupy the same waveform, bandwidth affects pulse compression, estimation precision, sidelobes, and radar image quality together with communication support [110], [111], [112], [113], [114]. Stimulated Brillouin scattering, frequency partitioning, and constellation choice reveal device and spectral limits [24], [115], [116]. Doppler

refinement depends on phase noise and computation [117]. A communication and sensing bandwidth ratio gains meaning from total bandwidth, target range, and receiver architecture [118]. These results show that signal bandwidth and receiver processing jointly define the operating point.

Reuse extends this mechanism by carrying sensing content in a shared waveform or an existing transport path [25], [119], [120]. Phase coding, carrier grouping, radio frequency bandwidth, and space and wavelength allocation then set the occupancy and isolation available to each function [51], [121], [122], [123]. The result depends on how the receiver separates the shared content and recovers the underlying sequence.

Hardware and processing resources complete the picture. Reference recovery and coding consume timing or redundancy [65], [124], [125], while emitter bandwidth, source coherence, optical budget, and distributed learning traffic set hardware and network capacity [126], [127], [128], [129]. Receiver count introduces alignment and crosstalk constraints. In the reported prototype, four surrounding receivers were designed and one receiver was operated [130]. Phase control and observation settings affect the recoverable path [8], [131], while modulation, sideband, and time bandwidth controls affect signal recovery [132], [133], [134]. The same nominal allocation can therefore produce a different result when the platform or receiver changes.

2) *Power, Energy, and Dynamic Range*: Power affects both functions through the finite operating range of the transmitter, channel, and receiver. Launch and received power, carrier and probe ratios, device bias, modulation swing, and converter headroom refer to different points in that chain. As power changes, performance follows the component that first approaches a noise, nonlinear, or saturation limit. The reference plane and limiting component are therefore part of the result.

A fiber probe sweep shows this response directly. Raising the sensing to telecom power ratio improves sensing SNR and raises BER, with the strongest link penalty near the probe [1]. In a visible light prototype, one optical branch supports communication and ranging while another supports energy harvesting. Their allocation governs rate, ranging quality, and harvested power under separate receiver and endpoint bias conditions [135]. Similar ratios connect BER or OSNR margin with phase, polarization, and vibration observability in fiber [3], [18], [136], [137], [138]. In photonic wireless links, carrier and waveform ratios connect phase recovery, BER or error vector magnitude (EVM), and radar or ranging quality within the available receiver headroom [32], [139], [140], [141]. Comparisons are most informative when they use the same power reference and receiver margin.

Absolute power sweeps reveal where the operating range ends. In fiber, launch and received power expose nonlinear interaction and sensing reach [21], [142], [143]. Photonic front ends examine operating limits associated with input power, modulator drive, converter linearity, and amplifier gain [38], [144], [145], [146]. Architecture then decides where the available power is used. Polarization paths, receiver taps, and branches assigned to individual functions distribute power before estimation [147], [148], [149]. Propagation, insertion, and alignment losses determine how much reaches those

branches [46], [67], [84]. The remaining margin sets the feasible operation of optical phased arrays, communication users, sensing estimators, and illumination services [68], [90], [150], [151].

Power allocation can also be written into the waveform. Service branches, reference copies, and relative waveform amplitudes connect radar masking, ranging, data delivery, and energy harvesting through a shared allocation [152], [153], [154], [155]. Sequence transitions, coding choices, and chaotic source parameters change how communication and sensing information is recovered [156], [157], [158]. Frequency separation, probabilistic shaping, sensing slots, and carrier reuse move energy and occupancy across signal dimensions [159], [160], [161], [162]. The waveform definition, hardware margin, and performance threshold determine the resulting operating point. These same factors also govern reliability and sensing quality.

3) *Communication Reliability and Sensing Quality*: Communication reliability and sensing quality become linked after the waveform and receiver are selected. Communication evidence uses BER, EVM, quality factor, and required OSNR, while sensing evidence uses SNR, estimation error, and detection. These measures can move together as leakage, nonlinear mixing, phase noise, cancellation quality, or receiver competition changes. Some controls reduce an interference path, while others redistribute energy within the shared signal. This distinction explains why similar metric pairs can follow different trends.

Two studies show why interference control and power redistribution require separate interpretation. In a fiber link combining communication with coherent reflectometry, sequence shaping reduces nonlinear interference induced by signal transitions. This lowers the communication penalty while retaining phase sensing at the reported condition [156]. In a millimeter wave waveform, a direct current offset strengthens the sensing component and raises the radar signal to interference ratio together with communication EVM [163]. The first mechanism changes how interference is produced, while the second redistributes energy within the signal. Their trends therefore arise from different controls.

Architecture then determines where these effects enter the chain. Shared path tests evaluate communication reliability beside sensing outcomes at stated operating points and validation settings [89], [164], [165], [166]. Dedicated monitoring wavelengths and reused channel estimates locate sensing support in optical filtering or receiver processing, with communication performance described qualitatively [33], [167]. The path and receiver therefore define the scope of each reliability result.

Once the architecture is fixed, waveform and inference choices determine the reported communication and sensing metrics. Carrier choice, direct current offset, and phase modulation affect recovery at stated settings [86], [168], [169]. Coding, guard structure, and waveform composition affect BER, correlation quality, and ranging error [170], [171], [172]. At the inference stage, loss weighting connects communication reliability with distance and velocity errors [173], while Kalman tuning acts within the sensing branch [174]. Shared sequence structure and finite block phase processing place

radar detection and phase estimation under their sample and noise models [39], [83]. Phase index and pulse position timing likewise connect communication error with radar SNR or time of arrival precision [175], [176]. The sample, noise, timing, and calibration definitions give each result its scope.

At the system level, access placement guided by channel sensing connects link capacity with weather and demand [88]. The aggregation of sensing traffic connects capacity with target count and resolution requirements [41]. Simultaneous use of time and frequency creates direct waveform coupling [177], while shared hardware with waveform switching creates a different path [178]. Waveform robustness under fading and learned phase sensing place the limiting response in the channel or processing stage [179], [180]. These studies connect reliability and sensing quality to the complete waveform, receiver, channel, and processing chain. The same chain also governs rate, resolution, accuracy, and range.

4) *Rate, Resolution, Accuracy, and Range:* The link between communication rate and sensing begins with the design variable that changes the rate. Bandwidth and symbol timing can alter resolution directly, while frame structure, geometry, and processing change the resources available to the estimator. The controlling variable and its operating conditions explain why a similar rate change can produce different sensing outcomes.

Waveform parameters provide the clearest starting point. In optoelectronic oscillator and remote ranging designs, code length and period connect bit rate or capacity with distance resolution and unambiguous range [181], [182]. Power division changes OFDM reliability, reach, and ranging error in a retroreflective prototype, while image decomposition links stripe recovery with scene reconstruction and recognition [19], [183]. LFM slope indexing and time division designs in the W band place their controlling choices in slope selection and frame duration [184], [185]. The shared parameter sets the direction of the relationship in each case.

Frame structure creates a second path by deciding when and where each function receives signal support. Dynamic allocation and intensity scaling distribute support across payload, sensing patterns, and coverage [2], [186]. Orthogonal time frequency space parameterization and complete grid OFDM reuse place both functions in a common time frequency structure [74], [187]. Separate measurement paths and mode switching show how a similar waveform can support different forms of integration [188], [189]. The frame and measurement schedule are therefore part of the result.

Geometry sets the region in which those resources remain useful. Steering and access point placement define communication and localization coverage, while surface allocation and blockage aware access select the paths available to each function [9], [78], [190], [191]. Source directivity and tag delay spacing further affect coverage, range separation, and concurrent return recovery [192], [193]. The reported layout is therefore part of the observed trend.

Signal construction can tie payload more closely to sensing observability. Chip length changes occupied symbol rate together with ranging precision and security tolerance, while embedded communication data affects distributed sensing fi-

delity [42], [194]. Amplitude shift keying rate and guard duration connect communication timing with range resolution and ambiguity [195], [196]. Waveform switching provides a related parameter coupling, with rate and resolution evaluated through separate parameter changes [197]. In the terahertz system, the available device band supports quadrature phase shift keying (QPSK) communication and the LFM bandwidth used to achieve range resolution [198]. The waveform and hardware band together define the feasible region.

Optimization studies express these mechanisms as operating regions. Energy and asymptotic block length define communication and detection rate pairs, while nonorthogonal multiple access power allocation connects aggregate rate with ranging error [50], [199]. Other studies combine capacity with ranging information or compare separate and joint operating configurations [77], [200]. Probabilistic shaping and embedded data create further regions through information rate, sidelobe level, and estimation error [201], [202]. The selected operating point follows the stated objective, constraints, and normalization.

Processing adds another layer because the estimator decides how much of the available signal becomes useful sensing information. Sampling period, spectral weighting, zero padding, and transform length exchange response time and computation for resolution under stated acquisition settings [203], [204], [205], [206]. Rate selection can tune the sensitivity of a density estimate derived from BER [207]. In a separate application, classifier size changes recognition accuracy and edge processing burden [62]. These studies distinguish waveform rate from the update and computation needed to interpret it.

Across the network, the relationship moves from a single waveform to the traffic and spectrum available across the system. Transport capacity controls how many sensing streams meet an echo quality threshold [208]. Separated bands, idle time frequency cells, cyclic prefix correlation, and shared optical waveforms connect occupied spectrum with resolution or estimation support [209], [210], [211], [212]. A visible light study carries this logic into a proposed capacity and localization problem [213]. The evidence thus distinguishes observed operation from a proposed design space.

5) *Waveform, Hardware, and Complexity:* Implementation complexity depends on where a design places its burden. Source coherence, receiver structure, calibration, synchronization, computation, memory, and latency represent different costs. Reusing a component may move work into processing or control, so implementation complexity must be assessed across the complete chain.

Two photonic systems show how this transfer occurs. A terahertz design uses coherent comb generation to stabilize the QPSK and LFM paths and concentrates complexity in the optical source chain [198]. A fiber experiment over a long span uses lower cost lasers with broad linewidths and recovers coherent data and vibration through residual carrier processing [214]. Each architecture preserves performance through a different balance of source hardware and digital processing.

6) *Other Implementation Relationships:* Other implementation relationships show how a shared choice appears at different layers of the system. Its effect may emerge as

protection, joint gain, device sensitivity, estimator burden, control freshness, or allocation latency. Each case carries its own conditions and prepares the comparison limits that follow.

Security and synergy illustrate two possible directions. Phase keyed recovery uses ranging tolerance to create a physically bounded decryption region [194]. With spatial alignment, communication gain and target profile estimation can benefit from the same optical design [68]. Their geometry, objective, and validation setting explain why one relationship expresses protection and the other expresses joint gain.

Other cases locate the relationship within the implementation chain. Electrode transparency connects device architecture with photodetector sensitivity, while passive return loss and repeater structure affect access for fault localization [60], [215]. Capacity and distance estimation define a model operating region [216]. Queue freshness enters the robotic control loop, and local spectral scanning connects computation with range resolution [47], [217]. Weighted priorities guide resource allocation, while orthogonal task channels shape application latency [218], [219]. Read together, these cases locate the relationship at the device, estimator, control, or allocation layer and provide the conditions needed for comparison across studies.

D. Conditions for Comparison Across Studies

Comparison across studies begins with alignment. The bounded relations summarized in Table V connect studies when their task, metric definition, measurement plane, operating point, and validation setting can be matched through the framework in Section II. Other metric evidence explains mechanisms, feasibility, and behavior within individual systems. This separation keeps each reported value attached to its original engineering meaning.

Tradeoff evidence follows the same principle. Because objectives and constraints vary across systems, the synthesis compares the shared mechanism and the conditions that produce the observed response. This approach shows why a relationship recurs and when it can guide another design. Its engineering value also depends on the validation setting and on how completely the configuration can be reconstructed. Section VI examines these requirements through validation settings, field evidence across both functional domains, and reconstructability.

VI. VALIDATION EVIDENCE, RECONSTRUCTABILITY, AND BENCHMARK READINESS

Section V showed that comparison depends on alignment across the task, measurement plane, operating point, and validation setting. This section examines how far those operating points were tested and how completely their configurations can be reconstructed. It then relates validation settings, evidence from both functions, and artifact access to benchmark readiness.

A. Validation Settings and Methods

Figure 9 places each study's highest reported setting beside the validation methods used across the corpus. The first panel assigns one setting to every study, while the second allows several methods to contribute to one evidence chain.

The profile shows a broad laboratory and prototype base together with a smaller field subset. Analytical, simulation, laboratory, and prototype methods can contribute to one study, so the configuration trail across methods shapes interpretation.

These settings expose different parts of an O-ISAC system. Simulation studies isolate waveform and device sensitivities under controlled conditions [85], [187]. Laboratory and prototype systems reveal conversion, synchronization, and receiver effects in physical implementations [60], [74], [110]. Field studies extend the evidence to access networks, deployed fiber, submarine monitoring, and outdoor wireless links [20], [33], [37], [38], [123].

The highest setting describes the reach of the evidence. Its scientific meaning still depends on the claim and the test design. A controlled prototype may explore a wider parameter range than a field trial, while a laboratory experiment may reveal communication and sensing interactions more directly than a field dataset collected for one outcome. The setting therefore gains meaning when it is read together with the tested conditions, available controls, and functional coverage.

Each method serves a different analytical purpose. Analysis clarifies assumptions, simulation supports controlled parameter studies, and laboratory work reveals the behavior of physical components. Prototypes bring subsystems together, while field studies introduce at least one operational boundary. Their combined value grows when the configuration and conditions remain traceable as the evidence moves from one method to another.

Section IV shows that realistic validation conditions change with the platform. Fiber studies depend on topology, traffic, launch and receiver conditions, and the disturbance context. Visible light communication and LiFi studies add layout, orientation, illumination, detector, and processing conditions. Free space optical studies require geometry, pointing, target properties, weather, and motion. Terahertz systems enabled by photonics must also trace optical conversion, antenna alignment, and wireless propagation. A useful validation record therefore identifies the environment, measurement plane, target or disturbance, reference method, and exercised function.

B. Field Evidence Across Communication and Sensing

The highest validation setting and the coverage of both functions answer related questions. For 12 studies, a field trial or deployment was the strongest reported setting. Six of them also received TQAF validation score 3 because their records report field or deployment outcomes in each functional domain. The assessment is applied to each study. Electronic Supplement S-Appraisal reports the corresponding scores. Electronic Supplement S7, the paired function validation register, identifies the 12 studies and the subset of six.

The six studies cover several settings. They include terahertz and cellular links enabled by photonics, fiber access

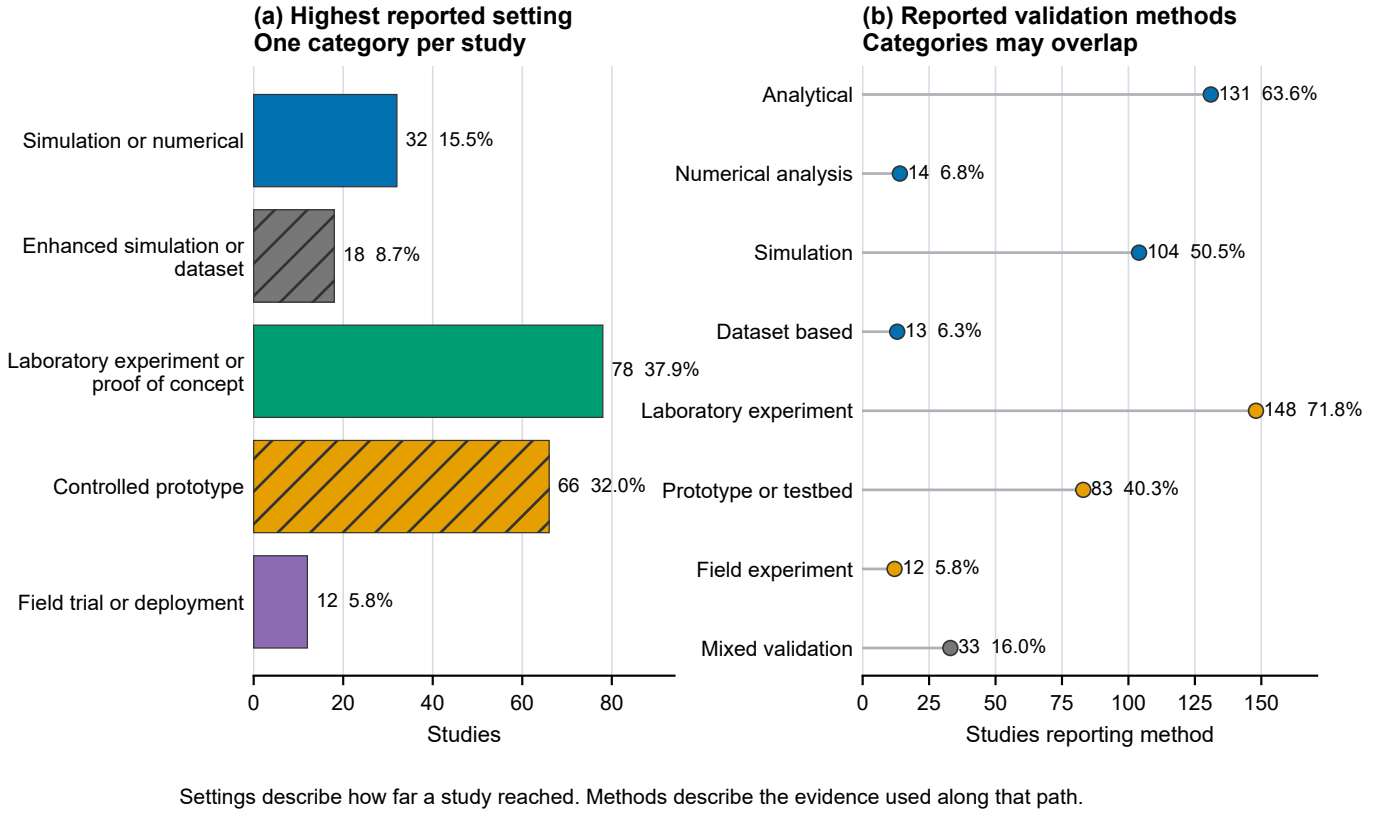


Fig. 9. Validation settings and methods in the 206 studies. Panel (a) assigns each study to one highest reported setting. Panel (b) reports analytical and numerical methods, simulations, methods based on datasets, laboratory experiments, prototypes, field studies, and mixed methods. Method categories may overlap because several forms of evidence can contribute to one study. The panels describe the reach and composition of validation as distinct properties.

integrated with radio frequency identification, and space laser ranging and communication. They also include multicore fiber deployed in the field and submarine cable monitoring [20], [33], [37], [38], [65], [123]. Their diversity shows that field evidence for both functional domains appears across several platforms. Platform comparisons still follow the native validation conditions described in Section IV.

Electronic Supplement S7 records the available relationship timing and separate source locators for the two functions. The six cases establish field or deployment outcomes in both functional domains at the level of each study. Concurrent joint operation is established when both outcomes are traceable to the same configuration, time interval, and condition set.

The distinction becomes important when evidence is produced at different stages of a system. One fiber study combines field data with offline sensing classification and illustrates how field collection and functional execution can occupy different parts of the evaluation path [120]. Segmented architectures add interfaces, timing, calibration, and conversion conditions to that path. When these elements connect both outcomes to one operating point, field coverage can support a stronger claim of joint operation.

C. Artifact Access and Reconstructability

Validation describes what a study tested. Reconstructability describes how completely another group can recover the operating point and the path from input to reported result.

TABLE VI
REPORTED ACCESS TO DATA AND TO CODE OR MODELS IN THE 206 STUDIES

Reported access	Studies, <i>n</i> (%)
<i>Data</i>	
Closed or status unreported	145 (70.4%)
Access by request	41 (19.9%)
Open access	13 (6.3%)
Outside artifact scope	7 (3.4%)
<i>Code or model</i>	
Closed or status unreported	197 (95.6%)
Access by request	7 (3.4%)
Partial components	1 (0.5%)
Outside artifact scope	1 (0.5%)

Note. The first row in each group combines closed access and an unspecified access status.

Access to data, code, or models supports this task, while physical experiments also depend on the system configuration, measurement conditions, processing steps, and provenance.

Table VI summarizes the access routes reported for data and for code or models in all 206 studies.

The reported routes support different forms of reconstruction. Open data appear across simulation, laboratory, prototype, computer vision, and field data settings [91], [110], [120], [187], [220]. They support reanalysis of the released data. Complete system reconstruction also requires configuration and execution details. Software preserves the transforma-

tion from inputs to reported outputs, and one computer vision study reported partial code or model components [91].

Reconstructing a physical experiment requires the system and hardware, the scene and calibration, the waveform and processing, the metrics and data, and the versioned release. These requirements extend the comparison profile in Section II. Electronic Supplements S-Data Dictionary and S-Evidence provide the field definitions and record provenance used in the review.

The TQAF reproducibility assessment reads artifact access together with parameter reporting. It places 4 studies in the low category, 199 in the adequate category, and 3 in the strong category. Electronic Supplement S-Appraisal provides scores for each study. Thirteen studies report open data, while three reach strong reproducibility. The difference reflects the additional reconstruction information captured by the appraisal.

D. Benchmark Readiness

A reconstructable result becomes a reusable benchmark when another group can apply the same task, condition set, baseline, and metric definition. The TQAF benchmark assessment combines these requirements with artifact access, validation, reproducibility, and comparison admissibility. The observed distribution comprises 48 studies in the low category, 158 in the adequate category, and zero in the strong category.

Under the review criteria, strong readiness depends on bringing an external or common baseline, an open artifact, admissible outcomes in both functional domains, and strong validation and reproducibility support into the same evidence chain. S-Appraisal reports these scores. This distribution directs attention toward shared tasks, baselines, artifacts, and evaluation conditions that other groups can reuse.

These distributions characterize evidence reach, functional coverage, artifact access, and reconstructability within the 206 included studies. They provide the descriptive basis for the benchmark and reporting actions in Section VIII.

Overall, the corpus combines a broad laboratory and prototype base with a smaller set of field and deployment results. Stronger cumulative evidence emerges when both functions are tied to one traceable configuration and the evaluation path is preserved through reusable artifacts. Section VII builds on this validation map by examining the technologies and applications supported by the reported evidence and their connection to 6G network roles.

VII. ENABLING TECHNOLOGIES, APPLICATION REQUIREMENTS, AND 6G NETWORK EVIDENCE

Validation shows how far a reported system was tested. We now follow the technologies that form communication and sensing observables into application requirements and 6G network roles. Because a study may contribute to several technology and application labels, their counts describe coverage within the corpus.

Figure 10 follows this evidence chain from optical generation and transport through waveform formation, spatial control, inference, selected application domains, and 6G network positioning.

The chain shows that network interpretation depends on continuity across the physical observable, the joint design choice, the application condition, and the reported network role. The displayed categories guide the reader through that relationship, while the following subsections retain the platform conditions and study evidence.

A. Technologies That Create and Shape Observables

The main technology labels occupy complementary points in the signal chain. Photonic generation creates or translates the carrier. Chirped processing maps delay or motion to beat frequency. Coherent reception preserves phase, and OFDM provides controllable time and frequency resources. A single architecture can combine several of these mechanisms.

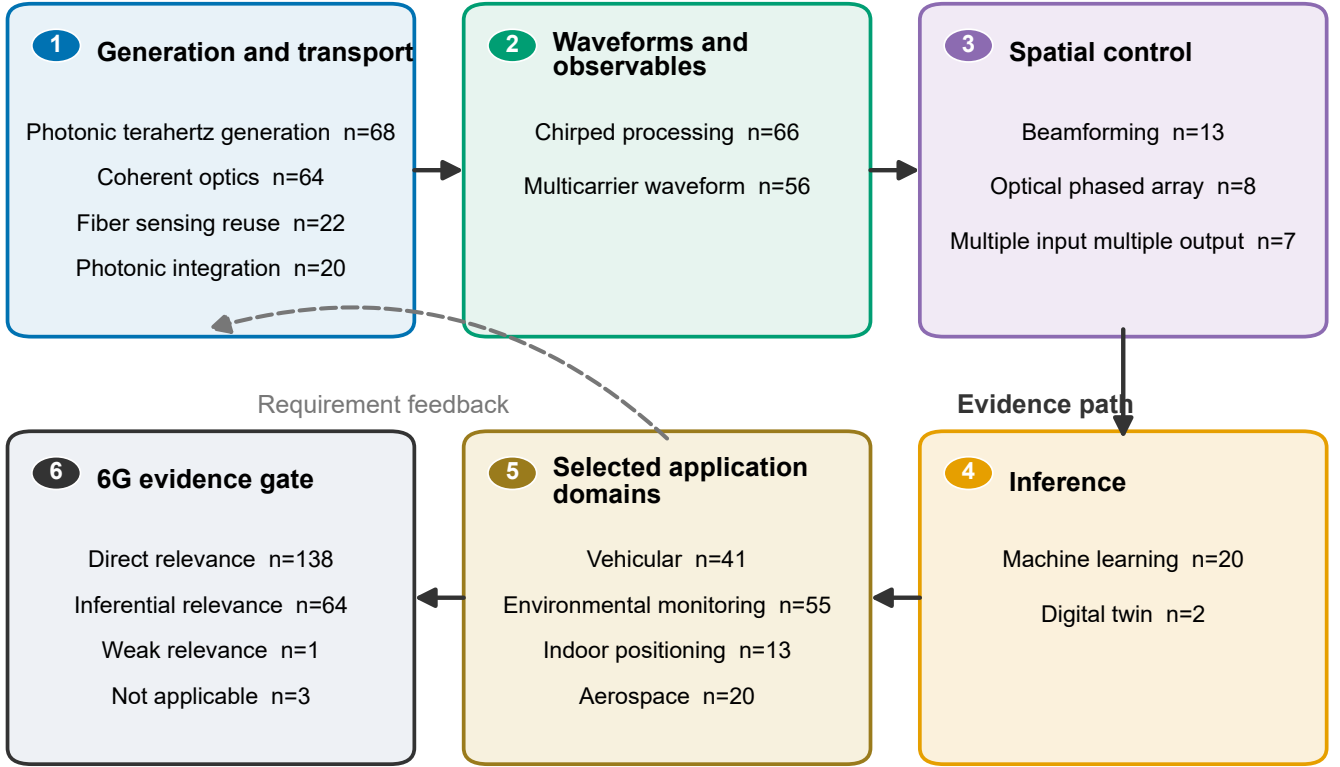
OFDM supports joint operation by placing payload and sensing resources within one signal structure. The sensing observable may come from pilots, payload tones, cyclic prefixes, reserved subcarriers, or a dedicated processing branch. This choice determines which samples are known at the receiver and how interference enters the estimate. Optical modulation constraints and the measurement plane used for SNR shape the interpretation. Embedded pilot fiber sensing and indoor optical positioning show how the same multicarrier principle serves different observables and geometries [85], [192].

Chirped and coherent systems draw on phase continuity and a shared timing reference. A returned chirp can reveal range or motion as the transmitted signal carries data. A coherent front end preserves small phase changes for distributed sensing and demodulation. Photonic generation can extend the reference to wideband millimeter wave or terahertz carriers. Full photonic dechirping and coherent fiber vibration sensing illustrate these two uses [42], [198]. Their performance depends on laser stability, reference distribution, conversion efficiency, receiver bandwidth, and calibration.

The optical stage governs carrier generation and reference stability. Once the carrier is radiated, the communication and sensing paths follow microwave, millimeter wave, or terahertz propagation. In a free space optical link, the optical beam itself traverses the channel. System evaluation should therefore follow the signal across both physical domains and identify where power, SNR, bandwidth, and timing are measured. This view makes the interface between components part of the joint system.

Spatial control changes both the communication link budget and the sensing geometry. Beamforming, optical phased arrays, multiple input multiple output processing, and reconfigurable optical intelligent surfaces provide different forms of spatial control. Beamforming directs energy. An optical phased array provides one steering mechanism. Multiple input multiple output processing uses several spatial dimensions, and an intelligent surface reshapes an intermediate propagation boundary. A system may combine these roles.

A narrow beam improves received power and angular selectivity while making acquisition, tracking, blockage, and calibration central to joint performance. The joint use of an optical phased array and OFDM and the use of frequency comb beamforming connect steering with ranging or imaging [68], [110]. Multiple input multiple output processing



Technology and application labels may overlap. The 6G evidence gate assigns one class per study.

Fig. 10. Organization across system layers of the mechanisms that create and shape O-ISAC observables and connect them to service requirements. Technology and application counts describe overlapping corpus coverage, so a study may contribute to several displayed categories. The 6G evidence gate is exclusive and assigns each of the 206 studies to one relevance class. Selected application domains illustrate the requirement flow, while Table VII provides the complete application inventory.

adds communication streams, sensing diversity, or separation among targets and paths. Structured fields and hybrid systems that combine free space optics with fiber Bragg gratings place these spatial dimensions at different points in the architecture [76], [81]. Meaningful comparison therefore requires the independent dimensions, geometry, and calibration method.

The intelligent surface evidence consists of one study that models a mobile free space optical path controlled by an optical intelligent surface [191]. A separate laboratory demonstration uses mechanically rotated Risley prisms to steer a photonic generated terahertz signal [109]. Both mechanisms redirect energy, and each uses different surface physics, propagation laws, control variables, and validation settings. Their separate treatment preserves the physical meaning and provides two starting points for broader experimental comparison.

B. Integration Beyond the Waveform

Integration and inference move joint design beyond waveform selection. Photonic integration appears in 20 studies, fiber distributed acoustic sensing in 22, and machine learning or artificial intelligence in 20. Two studies use digital twins. A

further 19 studies contain specialized mechanisms grouped as other because they fall outside the broad technology labels. Together, these categories cover device integration, infrastructure reuse, inference, and a diverse long tail.

Photonic integration can place generation, reference distribution, beam control, and detection on a compact hardware path. This arrangement reduces duplicated paths and brings packaging, thermal drift, crosstalk, and calibration into the same design problem. Microcomb synchronization coordinates wireless data with radar imaging. A distributed W band system uses radar estimates for beam alignment and flexible data service [221], [222]. Both examples show how a shared reference can connect communication and sensing hardware.

Fiber distributed acoustic sensing follows a complementary route by using the transmission medium as a spatial sensor. Phase, backscatter, and polarization can reveal events along the route as the fiber carries traffic. Coherent multicore transmission with distributed sensing and operational submarine cable monitoring illustrate two scales of this infrastructure reuse [33], [147]. The central design question becomes how to preserve the optical observable together with traffic continuity.

Learning turns optical observations into event labels, positions, field estimates, channel predictions, and resource deci-

sions. Each task is best read with its definition, reference data, training distribution, evaluation shift, uncertainty, and inference cost. Fiber diagnosis, federated visible light positioning, and wearable optical sensing show the range from distributed infrastructure to personal use [61], [120], [129].

A neural network can also take part in both functions. In one coherent link, the model compensates transmission impairments and estimates distributed power and abnormal loss [223]. The two outputs share the link while retaining separate targets and reference data. Digital twins add another model layer. One represents vehicular edge resources, and another provides a model aided physical reference for coherent fiber sensing [180], [219]. Clear reporting of the represented state, update process, mismatch model, and resulting decision gives these models an explicit system role.

C. Applications as Operating Requirements

Applications translate the mechanisms above into conditions that a joint service must satisfy. Table VII retains all thirteen application domains and maps each one to its joint role and defining evaluation conditions. A fiber platform for vehicles, for example, carries data at high rates and measures distributed temperature [224]. Devices used by people connect communication or remote alerts with motion, falls, air pressure, physiology, or gesture [61], [225], [226]. Each application therefore changes what the experiment must demonstrate, even when both functions share the same optical interface.

Across the map, the dominant requirement moves with the application. Network and mobility settings emphasize acquisition, timing, and continuity. Deployed infrastructure emphasizes persistent observation alongside carried traffic. Human use and extreme settings add physiological references, safety, privacy, pointing, atmosphere, packaging, and access to the test environment. These requirement bundles show which operating conditions must accompany a reported result before it can support comparison or reuse.

D. From 6G Relevance to Network Evidence

The 6G relevance gate in Figure 10 assigns each study one corpus position. Direct cases connect the reported system explicitly to a 6G role, as illustrated by [41], [110]. Inferential cases contribute an enabling link that gains meaning from additional network context, as illustrated by [42], [85]. The remaining classes capture a lighter connection to 6G or an application focus outside that framing, as illustrated by [167], [226].

The distribution connects the reviewed corpus with 6G access, transport, localization, automation, environmental awareness, and generation of carriers at high frequencies. A relevance code therefore locates a study within the 6G discussion. Standard conformance, interoperability, and deployment readiness require additional evidence tied to the relevant network framework [6].

A stronger network interpretation emerges when the optical or photonic subsystem occupies an explicit role across the complete network path. Access, fronthaul, transport, edge

processing, and service control bring timing, mobility, orchestration, energy, security, and recovery requirements. Evidence at this level evaluates useful communication and sensing under the conditions of that role and reports the resources needed to sustain both. This creates a traceable path from the physical observable and hardware to the service objective.

Within this survey, 6G relevance is strongest when both functional outcomes and their operating conditions remain traceable to a network role. The resulting evidence chain links physical observables, application requirements, and service objectives and provides the basis for the research priorities in Section VIII.

VIII. DISCUSSION, RESEARCH ROADMAP, AND LIMITATIONS

Across the evidence base, the central issue is whether physical sharing produces joint outcomes that remain traceable to one operating context. This section interprets that relationship, develops five research priorities, and states the review boundaries.

A. Interpretation Across the Evidence Base

Physical integration identifies where communication and sensing share infrastructure, signals, or information. Evidential integration shows how clearly both outcomes remain tied to the same operating conditions. Together, they connect architectural and transmission perspectives while preserving the physical regime of each platform [11], [12], [14], [16].

Deeper sharing can connect more design variables and improve coordination across the two functions. It can also couple power, distortion, synchronization, dynamic range, and failure behavior. Its scientific value therefore depends on locating these effects in the signal chain and tracing their influence on both outcomes.

Tradeoff patterns transfer across studies when tasks, metric definitions, measurement planes, baselines, constraints, and validation settings remain aligned. This alignment allows model bounds and empirical operating points to retain their distinct roles and support comparison within a stated context [15], [227].

Fielded cable and network studies illustrate the value of operational continuity [228], [229]. Within the survey, system evidence becomes reusable when configuration, calibration, processing, and reported artifacts remain connected across interfaces. This continuity links physical sharing to application and network claims and motivates a central research task. Joint operating points should remain interpretable, testable, and reusable across groups.

B. Research Priorities for Reusable Evidence

Table VIII organizes five priorities and pairs each observed finding with an action and a measurable sign of progress.

The order matters because each stage uses evidence produced by the previous one. Clear measurement context supports fair benchmarks. Benchmarks support realistic joint tests. Reconstructable artifacts make those tests repeatable,

TABLE VII
APPLICATION DOMAINS AND OPERATING REQUIREMENTS IN THE REVIEWED CORPUS

<i>Network and mobility</i>			
Domain, n (%)	Common communication and sensing role	Key operating condition	Studies
6G and access 100 (48.5%)	Access and transport with channel, target, position, or environment sensing	Timing, orchestration, continuity, and evidence for both functions	[41], [76]
Vehicular 41 (19.9%)	Vehicle data links with range, motion, pose, temperature, or channel sensing	Changing geometry, acquisition, tracking, and service continuity	[49], [224]
Optical access 32 (15.5%)	Access and transport links with channel, event, or fiber state sensing	Reach, synchronization, and traffic continuity	[41], [85], [120]
Indoor positioning 13 (6.3%)	Indoor optical links with user or robot position and orientation sensing	Geometry, calibration, coverage, and update continuity	[47], [192]
Data center 4 (1.9%)	Short reach data links with channel estimation or link state sensing	Traffic continuity and calibrated link references	[85]
<i>Infrastructure</i>			
Domain, n (%)	Common communication and sensing role	Key operating condition	Studies
Environmental monitoring 55 (26.7%)	Distributed links with vibration, temperature, or environmental event sensing	Persistent observation, localization, and traceable reference data	[33], [120], [206]
Smart infrastructure 41 (19.9%)	Infrastructure links with structural, motion, or event monitoring	Continuous observation, calibration, and resilient alert delivery	[76], [120], [126]
Industrial 34 (16.5%)	Industrial data or control with process, strain, temperature, motion, or position sensing	Delay, continuity, calibration, and safe operation	[61], [76]
Security and surveillance 26 (12.6%)	Monitoring links with anomaly, event, or location sensing	Coverage, event discrimination, and alert latency	[120]
<i>Human use, extreme settings, and remaining domains</i>			
Domain, n (%)	Common communication and sensing role	Key operating condition	Studies
Healthcare 6 (2.9%)	Wearable or ambient links with vital sign, activity, or gesture sensing	Motion robustness, physiological reference, illumination, and user safety	[60], [61]
Aerospace 20 (9.7%)	Satellite links with atmospheric or wavefront sensing	Pointing, turbulence, delay, and link adaptation	[73]
Underwater 2 (1.0%)	Underwater link concepts and subsea data links with temperature or event sensing	Water attenuation or cable calibration, packaging, and access	[126], [206]
Other 17 (8.3%)	Emerging links with range, imaging, object, or channel sensing	Metrics and reporting specific to the task	[109]

Note. Application domains overlap, so a study may contribute to more than one row. Counts and percentages describe coverage within the 206 included studies. For each domain, the table includes a selected study set and lists the corresponding references in the final column. The 6G and access row identifies the application category, and the final subsection provides the 6G relevance classification.

and repeatable tests provide a basis for adaptive and secure evaluation at network scale.

Stress tests should follow the native signal path of each modality. Fiber tests can vary topology and traffic, free space optical tests can vary geometry and atmosphere, and visible light tests can vary illumination and orientation. Photonic terahertz tests can vary optical drift together with wireless propagation. Matched baselines for the individual functions keep the limiting mechanism visible across these settings.

For standards and 6G coordination, the same sequence can support a core reporting profile that preserves modality physics, ground truth, and reusable benchmark definitions. Shared metadata can connect results across platforms while retaining the conditions needed for interpretation.

C. Scope and Limitations of the Review

The review represents English language technical reports published from 1 January 2020 through 22 June 2026 and indexed by the six searched sources. Full texts were available and assessed for 272 of the 330 reports sought, which corresponds to 82.4% retrieval coverage. Two Taylor & Francis exports with few records are documented through their com-

plete platform exports. These boundaries define the literature represented by the final corpus of 206 studies.

Searches and screening preceded retrospective registration, and the subsequent search refresh and report reconciliation produced the final corpus described in Section III. One investigator supervised routine decisions, while independent checks focused on specified evidence records and report reconciliation.

TQAF provides a technical appraisal developed for this review. The diversity of tasks, quantities, validation settings, outcomes, and source protocols supports structured narrative synthesis, which preserves native conditions and bounded comparisons. This analytical scale centers interpretation on reported mechanisms and conditions. Harmonized outcomes and source protocols form the appropriate basis for pooled effects, sensitivity estimates, and causal moderator analysis.

The multilabel taxonomy incorporates reviewer judgment, especially for hybrid systems. Row counts represent coded observations at extraction granularity, while effect analysis uses a different unit. For 92 metric rows, the available fields supported interpretation within the source setting and a conservative comparison status. Artifact access combined source reporting with targeted link checks, so the appraisal reflects

TABLE VIII
RESEARCH PRIORITIES DERIVED FROM THE REVIEWED EVIDENCE

Priority	Finding and proposed action	Evidence of progress
1. Measurement contract	<i>Finding.</i> Of 4,779 metric records, 118 supported bounded relations under aligned conditions. <i>Action.</i> Record the task, plane, unit, baseline, operating condition, validation setting, and source locator.	An independent reader reconstructs the quantity and condition from a structured record.
2. Benchmark for each modality	<i>Finding.</i> Comparison across studies depends on matched physical context. <i>Action.</i> Repeat one task with reference and independent designs under shared calibration and ground truth.	Independent groups reproduce the relative finding under the shared protocol.
3. Joint operation under disturbance	<i>Finding.</i> Of 402 substantive tradeoff relationships, 371 depend on reported conditions. Six of 12 field or deployment tier studies report outcomes in both domains. Traceability is strongest when both outcomes share the same timing and condition set. <i>Action.</i> Evaluate both functions in the same run and condition set under load, a disturbance relevant to the modality, and matched baselines for the individual functions.	Repeated trials recover the joint region and its limiting mechanism.
4. Reconstructable experiment package	<i>Finding.</i> Open data appear in 13 of 206 studies, and code or model access reaches a partial release in one. <i>Action.</i> Preserve a versioned package and rerun the main comparison under one added condition.	An independent rerun recovers the comparison and added condition from the package.
5. Intelligence, scale, and security	<i>Finding.</i> Technology and application coverage connects the corpus to 6G roles. Conformance, interoperability, and readiness become measurable through explicit network tests. <i>Action.</i> Apply a declared shift, attack, load, or component failure and report overhead, resource use, decisions, and fallback.	The service retains an explainable boundary with resource use, decisions, and fallback recorded.

Note. The findings summarize the retained corpus presented in Sections IV through VII. The proposed actions retain the physical conditions of each modality and define observable signs of progress.

documented reconstruction information.

The conclusions concern recurring mechanisms, measurement constraints, and validation patterns within the retained corpus. Corpus distributions describe the reviewed literature. Worldwide prevalence, market maturity, and platform rank require evidence designed for those questions. Within its stated scope, the review identifies the conditions that allow optical integration to support transferable joint results.

IX. CONCLUSION

Across the reviewed corpus, O-ISAC is organized most clearly by the physical role of optics and the mechanism that couples communication and sensing. Optics may carry the user link, illuminate the target, distribute a carrier, or preserve a coherent reference, and each role creates a different set of constraints. Shared hardware, carriers, waveforms, resources,

links, and design rules then connect the two functions at different points in the system. Their engineering value depends on how the task, metric definition, measurement plane, operating condition, and validation setting are reported together.

The evidence indicates that recurring tradeoff mechanisms can be compared when their physical context and measurement contracts align. Laboratory and prototype studies provide a broad experimental base, while field results illustrate how topology, geometry, propagation, calibration, and service continuity enter real systems. Application and 6G relevance become more informative when both functional outcomes remain traceable to the same network role and condition set. This makes transferable joint evidence the central measure of progress and keeps individual performance values tied to their source conditions.

Progress therefore depends on measurement contracts tailored to each modality, paired validation under realistic disturbances, and versioned artifacts that preserve configuration, calibration, data, and processing. These practices provide a path from individual demonstrations to reusable benchmarks and evaluation at network scale. They also allow subsequent designs to build on prior evidence while preserving the physical meaning of each optical platform.

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